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PART THREE
APPENDICES AND INDICES

APPENDIX A

ES-SUWEISSEH (SITE NO. 194) – A NEOLITHIC-CHALCOLITHIC SITE AT THE FOOT OF THE SAMARIA HILLS

Haim Winter

INTRODUCTION¹

The abundance of flint and stone tools dispersed densely all over the site justifies a preliminary publication of these tools. This study does not include all the technological, typological, and stylistic aspects of the tools, but attempts to document at least a part of these aspects for future comparisons to other sites.

Publications on contemporary sites in this region are rare; hence references are mostly to publications of excavated sites in the Shephela, the Negev, the Jordan Valley and the Golan Heights.

This study was carried out and completed in the late 90s, and published in Hebrew in 2000. It was translated into English in 2009 without any alterations; hence references to papers published since then are not included.

The flint finds

The sources of raw flint used at the site have not been investigated in detail. There were diverse kinds of flint present. For the larger artefacts: flakes, blades, celt tools, side scrapers, choppers, etc., coarse-grained, beige, grey-shaded flint was used. For smaller axes, sickle elements, and some other tools, good quality, brown and grey flint was used. Most blade-lets, retouched microliths, and other small tools were produced from high quality, mainly grey semi-translucent flint; but a few were of light brown, high quality flint. A few violet artefacts were apparently prepared from heat-treated cores (Nadel 1989). Charred tools were not found, but some charred waste materials were present.

The criteria for the identification of flint tools published in Volume 2

¹ Adam Zertal gave me the opportunity to study the interesting material from es-Suweisseh, the team of the Manasseh Hill Country Survey organized the visits at the site and the transportation of the artefacts, D. Eitam helped with the references, and Y. Winter identified the raw materials of the tools. I am grateful to all of them.

The drawings of the flint tools were prepared by me and the drawings of the stone tools by S. Haad.

(Winter 2008: 662-667) and the typological list for the Chalcolithic period (Gilead 1995: 276) with a few alterations according to local traits, served for the identification and presentation of this assemblage.

Debitage and waste materials

1. Primary flakes – $n = 24$ (10.13%): Any piece without secondary shaping and at least 30% cortex cover on its dorsal face is considered as a primary artifact (Goring-Morris 1987: 40; Rosen 1997: 31). We prefer this to the definition of 50% cortex (Gilead 1995: 276). Larger primary flakes were produced from coarse-grained flint, while the smaller pieces were detached from high quality, grey, flint cores. No primary elements in blade proportions were found.
2. Flakes – $n = 49$ (20.67%): Similarly to primary flakes, most large flakes (maximum dimension 8 cm, $n = 11$) were prepared from coarse-grained, beige, grey-shaded flint, while the smaller pieces were detached from high quality, grey, flint cores.
3. Blades – $n = 76$ (32.06%): In the assemblage, the relation of blade dimensions to the quality of the selected raw flint is clearly defined. Shorter pieces (6.5 cm maximum, $n = 35$. Some broken pieces were defined according to their shape and estimated length) were detached from high quality, grey flint cores, which were also found. The larger blades, length 7–15 cm, were mostly complete, and made of the porous, coarse-grained flint mentioned above. No such cores were present.
4. Bladelets – $n = 7$ (2.95%): Most bladelets (some broken), were prepared from grey, semi-translucent flint ($n = 6$). Cores of this flint variety were found.
5. Core trimming elements – $n = 1$ (0.04%): The single piece was made of brown, good quality flint.
6. Cores – $n = 22$ (9.28%):
 - a. This group ($n = 4$) includes quite large, light beige good quality flint. They served for the production of blades and flakes, and were not exhausted. Two were naviform shaped (Fig. A1:1).
 - b. Small, exhausted cores ($n = 16$) of grey semi-translucent flint with 1 or 2 striking platforms (Fig. A1:2), used for the preparation of bladelets and small flakes. Three pyramidal cores, well known from the Epi-Palaeolithic, were also found (Fig. A1:3).

The relatively high number of bladelet cores compared to the number of bladelets, was apparently caused by the fact that small artefacts are rarely discovered in a surface survey, as opposed to assemblages originating from excavated and sieved material. The absence of coarse-grained cores

used for the preparation of large flakes, large blades, celts, side scrapers, and some other tools present in the assemblage, is striking. These cores were apparently processed somewhere else, and only the semi-finished products were brought to the site.

7. Chips and chunks – $n = 58$ (24.47%): These partly charred artefacts were not classified according to the typological list. Only a few pieces of coarse-grained flint were present, thus supporting the thesis that these cores were processed elsewhere. Some violet pieces, apparently heat-treated, were also found.

Tools and retouched artefacts

1. End scrapers – $n = 13$ (6.99%): Various varieties of flint were used. The larger scrapers ($n = 7$; Fig. A1:4) were prepared from thick flakes of coarse-grained flint, and the abrupt scraping edge was shaped by a few crude scars. An additional group ($n = 5$; Fig. A1:5) were made of grey or brown high quality flint blades with fine, semi-abrupt retouch on the scraping edge. Two of these had scraping edges at both extremities, a phenomenon well known from the Epi-Palaeolithic. A single scraper was prepared from a truncated bilateral retouched blade of good quality beige flint, similar to that of the naviform cores.
2. Side scrapers – $n = 16$ (8.6%): All artefacts but two were prepared of coarse-grained, large (maximum dimension 12 cm) flakes. The active edge was mostly curved, with crude semi-abrupt retouch (Fig. A2:1).
3. Burins – $n = 10$ (5.38%): Eight of the tools were prepared on blades (Fig. A2:2), only a few being on coarse-grained flint. One burin was prepared on an unfinished bifacial celt. The smaller burins were of high quality grey flint, similarly to the small scrapers, bladelets, and microliths. Most tools had one longitudinal burin blow, and two had a transverse blow.
4. Awls and borers – $n = 4$ (2.15%): Barely represented in the assemblage, pieces found were prepared of coarse-grained flint. One short awl, on a primary flake, had bilateral abrupt retouch at the tip and clearly visible use-wear (Fig. A2:3). Another piece was an elongated borer prepared from a bifacial core tool. Two others were trihedral with abrupt retouch at the bit. Possibly these coarse tools were broken-off tips of picks. Three tiny awls on bladelets were also found – see the section on microliths below.
5. Retouched flakes – $n = 6$ (3.22%): The few large pieces were mostly prepared of coarse-grained flint. The retouch was semi-abrupt, dorsal, ventral or alternate; the length of the retouch was 2.5–3 cm.
6. Truncations – $n = 3$ (1.61%): Two truncated blades of coarse-grained flint

and one of good quality flint were found. All truncations were distal.

7. Notches – $n = 29$ (15.59%): This category was represented in assemblages of many periods, including the Chalcolithic. In our assemblage these tools were quite abundant – on flakes ($n = 16$) and on blades ($n = 13$). Small and shallow notches were usually on bladelets or flakes of grey or brown high quality flint, in some cases with additional retouch. The larger notches were prepared on large flakes or blades of coarse-grained flint (Fig. A2:4).
8. Denticulates – $n = 6$ (3.22%): All but one were prepared on large primary flakes or flakes of coarse-grained flint.
9. Sickles – $n = 10$ (5.37%): The quantity of sickles in the assemblage is rather poor, but nevertheless it included varieties which can be attributed to different periods. Six artefacts, some of them broken, were prepared from high quality, grey, brown, and violet flint. The violet piece had bilateral sheen and a fine serration on both cutting edges (Fig. A2:5). Similar artefacts were found in Jericho (Crowfoot-Payne 1983: 683), Yiftahel (Garfinkel 1987: 67), and Munhata (Gopher 1989a: 44-45). One grey piece had unilateral fine retouch and sheen. One segment of high quality, grey flint had fine, dorsal retouch on the cutting edge and abrupt retouch on the other edge. Sheen was visible on the cutting edge, the snap, and the truncation (Fig. A2:6). Two additional pieces, of similar flint and traits, showed sheen only on the cutting edge. Special attention should be paid to an elongated blade of violet flint, its distal end broken off, and its bulbar end thinned by ventral flat retouch. The edges were not retouched, but there was definite use sheen on the distal end of one edge. From its shape this was a hafted reaping knife (Fig. A2:7; Crowfoot-Payne 1983: 683; Gopher 1989a: 45). According to the colour and composition of the pieces it may be assumed that some of the artefacts were detached from heat-treated cores (Nadel 1989). Despite the small sample, these segments should be attributed to the Pre-Pottery Neolithic B period (Crowfoot-Payne 1983; Gopher 1989a; 1989b; Goring-Morris 1991).

Two segments of good quality light brown flint were also found. Both had a trapezoidal cross-section (Rosen 1997: 44-45) and some patches of cortex on the dorsal face. Each had one snap and one truncation on the extremities, and abrupt backing. The cutting edges showed fine dorsal serration and sheen along the cutting edge (Fig. A3:1). These segments should be attributed to the Chalcolithic period (Jericho – Hennessey 1969: 18; Besor – Roshwalb 1981: 29; Shiqmim – Levy and Rosen 1987: 288;

Beth Netofa – Gilead 1989: 265; Grar – Gilead 1995: 25; Rosen 1997: 50).

The third variety was a single sickle segment from beige to grey coarse-grained flint with a trapezoidal cross-section, two dominant dorsal ridges, use-wear, and sheen on one edge. These tools were defined as Canaanean blades attributed to the Early and Middle Bronze Ages (Neuville 1930; Rosen 1983: 127; 1989: 208; 1997: 16-48). Apparently this is an artifact from a later period, not related to the main assemblage.²

10. Retouched blades – $n = 25$ (13.44%): These are subdivided into three groups:

- a. Short blades of brown or grey high quality flint ($n = 14$). Most had fine dorsal retouch, and in some cases the whole length was retouched. Few had ventral retouch. The width after the retouch exceeded 9 mm; therefore they were not included in the bladelet category (Tixier 1963; Goring-Morris 1987: 48).
- b. Longer and wider blades of porous coarse-grained, beige tending to grey flint ($n = 8$; Fig. A3:2). All had one edge with abrupt retouch, and some also had flat ventral retouch on the other edge. None of the artefacts had any sheen, and they should be defined as backed knives.
- c. Similar pieces of coarse-grained flint ($n = 3$; Fig. A3:3) with flat, either dorsal or ventral retouch on one edge. These pieces were apparently also cutting tools.

11. Microliths ($n = 12$, 6.45%): The small collection of retouched microliths included several types: retouched bladelets ($n = 2$); abruptly backed bladelets ($n = 4$; Fig. A3:4); micro endscrapers ($n = 2$; Fig. A3:5); awls on bladelets ($n = 3$; Fig. A3:6); and a microburin (included in the tool group only for the discussion) ($n = 1$; Fig. A3:7).

All the microliths were prepared from high quality, grey, semi-translucent or brown flint. The assemblage also contained cores matching the artefacts.

12. Celt tools ($n = 27$, 14.51%) found were: bifacial axes ($n = 14$), bifacial adzes ($n = 10$), and trihedral picks ($n = 3$). Here also the choice of the flint variety matched the size of the tool. For the larger axes and adzes coarse-grained flint was used, while for smaller tools, high quality grey or light brown flint was used. No polished celt tools were discovered. All the axes had a lentoid cross-section, and on some the active end was shaped by a transverse blow (Fig. A3:9). The larger pieces ($n = 2$, 14%)

² Studies conducted after this chapter had been completed (Bar and Winter 2010) show that the Canaanean technology was already used as early as the Chalcolithic, so this single artefact could also be included in the regular assemblage.

were prepared from coarse-grained flint, and the smaller ones ($n = 12$, 86%) from good quality flint. The adzes had a trapezoidal cross-section and a flaked ventral face (Fig. A3:10). The larger pieces ($n = 8$, 80%) were of coarse-grained flint: two other (20%), very small pieces were of good quality flint. The picks are large trihedral tools of coarse-grained flint (Fig. A4:1). The absence of chisels and polished tools is noteworthy.

13. Choppers – $n = 2$ (1.07%): The two choppers were of coarse-grained porous flint. The faces were prepared by removing some of the cortex by flaking, but larger patches of cortex still remained (Fig. A4:2).
14. Arrow heads – $n = 1$ (0.53%): A single Byblos type arrow head, of high quality grey flint with a retouched shaft, was found. Both lateral edges of the tip were shaped by semi-abrupt dorsal retouch (Fig. A4:3).
15. Varia – $n = 22$ (11.82%):
 - One transverse scraper on a primary flake with a large notch.
 - One 11 cm-long blade of coarse-grained flint, trapezoidal cross-section, and abrupt bilateral dorsal retouch. On one lateral edge was an elongated (4.5 cm) ventrally retouched notch.
 - One point on a 12 cm-long blade of coarse-grained flint. Both lateral edges had abrupt dorsal and flat ventral retouch (Fig. A4:4). Similar points were attributed to the borer class (Mallon *et al.* 1934: fig. 19; Rosen 1987: fig. 22.10:19).
 - One tool, prepared from a large coarse-grained flint flake. It was ovoid, resembling a hand axe (biface), with patches of cortex on both faces. The retouch was bifacial, semi-abrupt, and crude. Definite signs of hammering were visible around the circumference. Because of the bifacial retouch and the marks of hammering, the tool cannot be defined as a rounded scraper, even though its shape was similar to a tool reported from Shiqmim (Levy and Rosen 1987: 579, fig. 10.7).
 - Six small artifacts of fine, grey, brown, and violet flint with marginal retouch. None of these can be classified according to a typological list.
 - 12 hammer-stones, seven of which were made on coarse-grained flint cobbles with hammering scars, at least on one end. One of these was apparently prepared on an unfinished or broken axe or adze, with bifacial modification. Four smaller examples were of high quality grey flint. Two of these were globular with hammering marks all over the surface. Hammer-stones were included in the tool category because they actually served as tools.

***Rock-cut installations; vessels of basalt,
limestone, and rare stone***

At the site many installations of different shapes were cut into bedrock surfaces and rocks. A detailed study of these installations is beyond the scope of this Appendix, but without doubt, the stone tools found were used in them. Thus these installations should also be reviewed, at least in brief. A study combining installations and tools could illuminate the way of life, daily activities, and even nutritional habits of the palaeo-population. In this study stone tools will be described in detail without analyzing their functional aspects.

Rock-cut installations

The survey discovered scores of installations which can be divided into categories according to their shapes and dimensions. In some cases, installations were concentrated close to each other in a group on a limited space.

1. Small cup-marks with a conical cross-section, 8–12 cm diameter (Fig. A5:1a), concentrated in groups of 2–20 cup-marks (Eitam 1999: 119). They were present both on flat bedrock surfaces, and on blocks removed by road construction. In some cases they were found near large basins cut in the bedrock. In a few cases a pair was connected by a shallow groove.
2. Round mortars. The rims were quite sharp, and the cylindrical parts almost parallel-sided (Fig. A5:1b). They were 18–42 cm in diameter and 17–40 cm deep.
3. Large, asymmetric basins. The rim was usually quite sharp, and the bottom flattened (Fig. A5:1c). Some of these basins were damaged by later activities. Two typical basins were 48×60 cm, depth 22 cm, and 47×60 cm, depth 20 cm. One basin was cut into the bedrock and damaged an earlier group of cup-marks. This indicates a long-term use, or even a multiple strata site.
4. Plate-shaped asymmetric basins, 13–15 cm diameter and 2–3 cm deep, with small cup-marks and round mortars, mostly of fairly equal diameter and depth – 4–5 cm, in the centre (Fig. A5:1d).

Stone tools

Quite a large variety of stone tools was collected at the site. Since the functional aspects of the tools have not yet been studied in detail, the classification is mainly according to their morphology. The name of the group in this study usually follows the generally accepted terminology, not necessarily matching the actual function of the tool. For example: the elongated

cylindrical tool defined as 'pestle' could easily be used as a rolling-pin, and not in all cases it was used for pounding in a mortar.

Generally, the tools may be divided into the following groups: containers; pounding and crushing tools; milling and grinding tools; sharpening and polishing tools; diverse undefined tools. The differences between milling and grinding have been studied by Avitsur (1976: 71-76) and Eitam (1999: 119-121).

The site is located in a ploughed agricultural area; therefore only a few unbroken artefacts were found. Most stone tools were of basalt of various qualities. The minority were of limestone and other, still unidentified, types of stone.

1. Containers – $n = 18$ (18.95%): In this study any vessel with a cavity was defined as a container. Most finds were fragments of bases or rims which allow the identification of the original shape of the artifact. Hence the list of finds:

- Four wall fragments of deep conical bowls.
- Two rounded rim fragments of similar bowls, clearly indicating the conical shape of the tool (Fig. A5:2).
- Five flat bases. The remnants of the walls showed the original conical shape of the bowl. One base fragment was surrounded by a protruding ring (Fig. A5:3).

These fragments apparently belong to the same type of bowl, made of homogeneous, fine-grained basalt. The walls were 1–2 cm thick. According to one bowl rim fragment the original diameter was 31 cm. Two bases were 15 and 18 cm in diameter. No decorations were found on the rims.

- Five basalt fragments of round pedestal bases (Fig. A5:4). One complete conical bowl on three or four pillars connected to a circular base was found. Similar vessels were also produced from clay, and defined as incense burners (Amiran 1969: 24, 302-303). Possibly some of these conical bowl fragments belong to these objects.
- One basalt mortar fragment, 12 cm diameter and wall thickness about 2.5 cm (Fig. A5:5).
- One limestone mortar fragment, 16 cm diameter and wall thickness about 2.5 cm (Fig. A5:6).
- One small, shallow bowl on a pebble of calcareous flint with a cavity 6.5 cm diameter and 1.5 cm deep (Fig. A5:7). The bottom of the cavity had a red colouring that could not be removed by washing, apparently remnants of ochre.

2. Pounding and crushing tools – $n = 33$ (34.74%): This group includes cylindrical elongated tools generally defined as pestles. Their distal end was used to crush the processed material into a mortar. Wear signs should be expected at the distal ends of the tools. As, so far, no detailed study concerning the application of the tools present in the assemblage had been carried out. All tools of similar shape were included in the category of pestles. On many items, use-wear signs were also present on the circumference of the tool, and not only on the ends as expected. These signs included polish and sheen, pointing apparently to the processing of grain. The tools can be divided into sub-categories according to their shape and the location of the wear signs. Despite the fact that many tools were broken, most of them could be identified:
 - a. Conical elongated pestle – $n = 11$ (Fig. A6:1): About half of these tools were short cylindrical fragments. The cross-sections were either round or nearly round, with diameters of about 5 cm. Their original length cannot be estimated. In most cases only the thinner part was preserved. The complete tools were 13–23 cm long and 3.5–6.5 cm in diameter. Eight pestles were of basalt, and three of fine-grained dense limestone or dolomite. On the complete pieces it was observed that the thicker end of the tools was barely marked, as this was not the active end of the tool. The thinner end did not show any pounding signs, but its circumference was polished, thus proving that this was the active part of the tool. Apparently these tools served for crushing or milling grain, or possibly other foodstuff, in the conical basalt bowls (Fig. A5:2).
 - b. Cylindrical barrel-shaped pestle – $n = 16$ (fig. A6:2): Nine of these tools were complete or nearly complete, and seven were short fragments. The complete tools were 10–17 cm long and 3.5–7 cm in diameter. One was of calcareous flint, one of limestone, and the other 14 of basalt. The cross-section of these tools was almost perfectly round, and they were cylindrical or slightly barrel-shaped. There were no signs of polish or sheen, created by rubbing, similar to that found on the conical pestles. This fact proves that the tools served mainly for rolling on a straight or slightly curved surface, and not rubbing. Both ends of the tools were, after prolonged use, slightly rounded. The tools were apparently used for grinding or crushing of still unknown materials.
 - c. Pounding pestle – $n = 6$ (Fig. A6:3): These tools had diverse cross-sections: rhomboid, triangular, elliptical, round, and irregular. On some of them faint polish was visible on the shaft. One end was rounded, proving that this was the active part of the tool. These tools were used

for crushing and pounding in the mortars present at the site, some of which were cut into the bedrock (Fig. A5:5, 6). Two nearly complete pestles, 17 cm long, and one fragment, were of limestone, while the other three were of basalt.

3. Milling and grinding tools – $n = 19$ (20.0%): The characteristic trait of these tools was that they had at least one slightly convex polished surface. This category was subdivided according to morphology and type of stone:

- a. Axe-shaped grinding stone – $n = 8$ (Fig. A6:4): At a first sight the impression was misleading, and the tool could have been mistaken for a basalt axe. The tool had two convex surfaces merging into a sharp round edge, similar to the cutting edge of an axe. No use scars similar to those of axes were found on these edges. The cross-section of the opposite end was round or slightly oval. The surfaces and the sharp edges were polished. All these tools were of fine-grained non-porous basalt. Axe-shaped basalt tools are known from the Neolithic, and occasionally they are regarded as cult objects. The Neolithic tools, except for the polished sharp edge, do not resemble the tools in our assemblage. These artefacts were prepared from stone which is too soft for cutting harder materials, thus they were not classified as axes. An experiment that we conducted at es-Suweisseh proved that the shape of the tools, both the length and the curved surfaces, perfectly matched the shape of the small cup-marks with a conical cross-section and a concave bottom (Fig. A5:1a). Accordingly, the tools were defined as grinding tools used in these cup-marks. The polished curved surfaces and the sharp edges were constantly shaped during the grinding process; hence there were no cutting scars on the edges. The opposite ends were prepared by detaching a few flakes in order to prepare an easy handhold. On some of the tools the polished concave surfaces were not yet complete, thus confirming that they were shaped during prolonged use. Three complete tools and five broken pieces with complete sharp edges were found.

- b. Discoid grinding stone – $n = 8$ (Fig. A6:5): All these discoid tools had at least one typical smoothed flat surface; otherwise they could have been defined as covers. Four complete tools and one fragment were of limestone, with diameters of 7–11.5 cm, and the other four broken pieces were from basalt.

- c. Grinding stones of various shapes – $n = 11$:

- Four tools made of un-worked calcareous flint cobbles, one of which was broken. The polished smoothed surfaces indicated the function

of the tools. The lengths, 6–12 cm, provide an easy handhold. On one of the pieces residues of red ochre were observed.

- Five partly broken basalt artefacts, with natural or prepared handholds. All had polished smoothed surfaces and the measurable length was 8–10 cm.
- One complete loaf-shaped tool, 9×11.5 cm, thickness 7 cm (Fig. A6:6).
- One fragment of sandstone.

4. Sharpening and polishing tools – n = 8 (8.42%):

- a. Point-sharpening tools – n = 5 (Fig. A6:7): Characterized by a longitudinal rounded groove of various depths, they were prepared from elongated cobbles with an easy handhold. Two complete tools: length 10 cm, width 5.5–7, cm, and thickness 3.5–5 cm, were prepared from hard limestone. The depth of the grooves varied from 0.3 to 1 cm, and the width from 1 to 2 cm. Some basalt pieces were apparently prepared from discarded pestles.
- b. Polishing and burnishing tools – n = 3: All pieces were of grey fine-grained limestone or dolomite. The surface, except for scars and breaks, was burnished. The cross-section was quite irregular, resembling a biconvex lens. The two larger pieces were broken. The dimensions of the first were 11×5.5×3.5 cm, and of the second were 16×6.5×3 cm. The third, complete, artifact measured 10.5×3×1.5 cm. The tools were defined as polishing and burnishing tools because of their burnished surface: perhaps they were used in the production of ceramic vessels (Amiran 1971: 22–23).

5. Various undefined tools – n = 17 (17.89%): All the tools which could not be classified according to the above categories, tools with undefined functions, and fragments with use-wear signs were included in this group.

- One basalt adze with a trapezoidal cross-section and a broken cutting edge. The tool was shaped by coarse flaking, similarly to the adzes of coarse-grained flint described in the section on flint tools. The remaining length was 18 cm, the width 5 cm, and the thickness 3.5 cm.
- One round polished basalt fragment, estimated diameter 16 cm, with a partly broken conical hole 2.5–4.5 cm in diameter. This hole was carefully polished by a continuous rotational motion. Similar tools were found at Metzger, and apparently also at Horvat Beter (Dothan 1958: 27; 1959: 28), and defined as parts of a potter's wheel. The dimensions of our tool are similar to those of the complete artifact at Metzger.
- Two short cylindrical basalt artefacts, one of them broken, with a

round depression in one end (Fig. A6:8). The dimensions of the complete tool were 3.8–4.5 cm diameter, length 5.5 cm, and depth of the depression 1 cm.

- One slightly concave basalt fragment of a lower grinding slab (Fig. A6:9). The dimensions were 10×10 cm, and thickness 4–5.5 cm.
- Two basalt discs, 5.5 and 8 cm in diameter and a thickness of 2.5 cm, possibly plugs.
- Three polished fragments of phosphorite. Artefacts of this material have been found at various Chalcolithic sites (Goren 1991: 102–111; Gilead 1995: 314–318).
- Seven undefined, worked basalt fragments.

DISCUSSION

The scope of this study does not allow a comparison to all relevant data from scores of sites; hence we concentrated on the publications from the following sites: Teleilat Ghassul (Mallon *et al.* 1934; Hennessey 1969), Abu Matar (Perrot 1955), Metzer (Dothan 1957; 1959), Horvat Beter (Dothan 1958), Neve Ur (Perrot *et al.* 1967), Shiqmim (Levy and Rosen 1987), Grar (Gilead 1995), and the Golan (Epstein 1998).

The flint tool assemblage contained a wide range of raw flint qualities. Various materials were used for categories differing in size, workmanship, technology, and apparently also in their application.

A large part of the assemblage was prepared from coarse-grained, beige tending to grey, flint. This variety was used for the larger tools: pestles, picks, adzes, axes, side scrapers, and others. The retouch was prepared by coarse flaking, and few examples of fine retouch were found. An investigation of the relation of the coarse flaking to the properties of the material is beyond the scope of this study. Such an investigation needs a full technological analysis, including experiments. It should be noted that no cores of this flint quality were found. Different, good quality, beige and brown flint was used for the preparation of a scraper, axes, and some other artefacts. Quite a few tools of this material were smaller than the previous group: the retouch fine, flat, semi-abrupt, and abrupt. Cores from this material, including naviform cores, were present.

A third variety of mainly grey, high quality flint was used for the production of axes, burins, scrapers, and microliths; the tools were mostly small. Cores and waste of this material were also found. Apparently the rarer materials were fully exploited, a phenomenon also reported from other

sites (Grar – Gilead 1995: 274).

There could be several reasons for the use of a variety of raw materials:

- a. The material was chosen according to the function of the tool;
- b. A time gap in the production of the tools within the same cultural entity;
- c. Tools of different raw material originated from different entities which occupied the site at different times. For example – the Pre-Pottery Neolithic and the Chalcolithic;
- d. A typological analysis could confirm a combination of both possibilities a and d.

The small, pyramidal cores, the various microliths, the backed bladelets, and the double-ended scrapers apparently indicate an occupation during the Epi-Palaeolithic or the Early Neolithic. Bladelets and scrapers could be attributed to the stages of the Kebaran or Natufian; the awls on bladelets to the Natufian and Early Neolithic (Netiv Hagdud – Nadel 1988; Hatula – Ronen *et al.* 1994: 181; Winter and Ronen 1994: 259); and the microburin mainly to the Natufian. The quantity of these artefacts was rather small compared to Epi-Palaeolithic sites, and most of them were also present in Chalcolithic sites (Rosen 1997: 67); accordingly, there is not enough evidence for an occupation during the Epi-Palaeolithic.

The smaller part of the assemblage should be attributed to the Pre-Pottery Neolithic B period. This group includes:

- a. Cores from brown-beige good quality flint, including naviform cores. Such cores were present at Jericho (Crowfoot-Payne 1983: 667) and Yiftahel (Garfinkel 1987);
- b. Axes with a bi-convex cross-section and cutting edges shaped by transverse blows;
- c. Burins;
- d. Sickles, especially those from heat-treated flint (Nadel 1989);
- e. The reaping knife (Gopher 1989a: 45);
- f. The Byblos type arrowhead (Gopher 1985: 51).

Most axes of the assemblage (86%) were small, prepared from good quality flint, and the cutting edge was prepared by transverse blows. Such axes were found at many Neolithic sites (e.g. Jericho – Crowfoot-Payne 1983: 631, 675) but they also were present at some Chalcolithic sites (Shiqmim – Levy and Rosen 1987; Grar – Gilead 1995). The use of the transverse blow for shaping the cutting edge of axes began during the Neolithic (Crowfoot-Payne 1983: 675). During the Chalcolithic it was apparently applied only for

resharpening blunted cutting edges (Perrot *et al.* 1967: 217; Bar-Yosef *et al.* 1991: 412). It should be noted that it is assumed that burins were sporadically present in post-Neolithic assemblages (Rosen 1997: 100).³

All the above-mentioned artefacts were from good, beige, brown, and grey raw material. The workmanship and the retouch are at a good level. This part of the assemblage apparently confirms the occupation of the site during the Pre-Pottery Neolithic B. The quantity of the artefacts and the composition of the assemblage exclude the possibility that they were collected elsewhere and brought later to the site.

In our opinion, the main bulk of the assemblage should be attributed to the Chalcolithic period. According to studies from Teleilat Ghassul (Mallon *et al.* 1934; Hennessey 1969; Lee 1973), Shiqmim (Levy and Rosen 1987), Besor sites (Roshwalb 1981), Beth Netofa sites (Gilead 1989), and Grar (Gilead 1995) during this period, the tool assemblage included adzes, axes, chisels, picks, end and side-scrapers, sickle blades, and various microlithic tools. Microliths and pyramidal cores from grey semi-translucent flint were also found at Chalcolithic sites (Besor sites – Roshwalb 1981: 58, 348), where backed bladelets were also present (Roshwalb 1981: 92, 375; Rosen 1997: 65), but the relative share of microliths in Chalcolithic assemblages varied considerably (Gilead 1995: 475). The micro end-scraper was typical of the Chalcolithic (Gilead 1984), and so were the awls on bladelets (Gaza M – Roshwalb 1981: 188). Adzes with a trapezoidal cross-section were also characteristic of this period. In our assemblage these were large, up to 18 cm in length, and 80% were from coarse-grained flint. Only 14 % of the axes with a bi-convex cross-section were prepared of this coarse-grained material. They were larger than those prepared from good quality flint, and shaped by coarse flaking similar to the adzes, picks, and other coarse tools. They obviously belong to the same group of tools. Accordingly, the majority of the flint assemblage of es-Suweisseh, apart from the artefacts attributed to the Neolithic, originated from the Chalcolithic. The fact that some types were also present during other periods does not contradict this fact. Even though *fossile directeur* pieces like perforated discs, rounded and fan-scrapers were not found, this does not contradict our dating. These tools were only found in some contemporary sites, and were apparently distributed by trade networks (Levy and Rosen 1987: 286; Rosen 1997: 75). The large side-scrapers of coarse-grained flint (as was most of the assemblage) with a curved working edge, perhaps served the same purpose as conventional fan-scrapers. A remarkable phenomenon is the absence of

³ Later studies proved that burins were present also in post-Neolithic assemblages.

chisels and polished tools. Polishing of tools started in the Neolithic, and was also common in the Chalcolithic. At contemporary sites in the vicinity, Metzger (Dothan 1957: 226) and Tel Esur (E. Yannai, personal information), such polished tools were present. At our site the tools were possibly overlooked casually in the course of the surface collection, but as in es-Suweisseh and also at Grar, they were very rare (Gilead 1995: 258-259). The possibility of whether the presence of polished tools has some chronological implications within the Chalcolithic should be checked.

The function of the flint tools

The character of the activities in a certain period affects the size and morphology of the tools and the choice of the raw material for their production. According to their shape the large adzes could have served as hoes, and together with the picks they could be evidence of agricultural activities. The analysis of the assemblage of es-Suweisseh proves that during the Chalcolithic period a variety of raw materials was used for the production of a diversified tool kit intended for a range of activities. High quality flint was used for microlithic tools, and coarse-grained flint for large tools. It is as yet impossible to define whether this choice was made according to the function of the tool (version a. above), or changes in the mode of flint working took place (version b. above) during the period.

The finds from the Neolithic prove that materials different from the Chalcolithic were chosen, good quality flint was preferred, and no coarse-grained flint was used (according to version c. above).

Rock-cut installations

Rock-cut installations from the Chalcolithic period, similar to those of es-Suweisseh, were published from Metzger (Dothan 1959: 227) and the multi-period site Khallet el-Gazaz and vicinity (Eitam 1993: 70-83; see also Site 114). A detailed study of the installations at our site, which was not occupied later on, could shed new light on the special purpose of these during the period.

At the site many installations of different shapes were hewn into the bedrock surfaces and the rocks dispersed at the site. A detailed study of these installations has not yet been carried out, but beyond any doubt, the stone tools found were applied in them. Thus, these installations should also be presented at least briefly (Fig. A5). A study combining installations and tools could illuminate life ways, daily activities and even nutrition habits of the palaeo-population.

Stone tools

The above order of presentation of stone tools and installations will be used henceforth.

Basalt, limestone, and phosphorite vessels were common during the Chalcolithic period (Amiran and Porath 1984: 11; Gilead 1995: 319). Limestone and dolomite outcrops are located close to the site on the slopes of the hills, and pebbles are found in the wadi beds. During the period these were widely used for the production of tools (Hennessey 1969: 19; Gilead 1995: 321). Basalt and phosphorite, not found in the surroundings, were brought from elsewhere. Similarly to other sites, some of the vessels were not produced locally, but distributed by trade networks and imported (Perrot 1955: 78; Gilead 1995: 318). The utilization of phosphorite for the production of tools was discovered by Gilead and Goren (1989), an apparently unique phenomenon of this period. At first, due to its colour, phosphorite was wrongly classified as basalt. Only petrographic analyses, which are beyond the scope of this study, can identify the source of the materials.

The conical bowl of basalt or phosphorite (Fig. A5:2) was defined as an offering vessel typical of the period. It has a flat base (Fig. A5:3) or a ring base with 3 or 4 supporting pillars (Fig. A5:4). The use of these items is still being discussed. Some scholars assume that these were ordinary household utensils for daily use (Gilead 1995: 318-322), while others assume that they were precious cult objects (Amiran and Porath 1984: 134). Fragments of these objects were found in nearly all Chalcolithic sites. Complete bowls were reported from Teleilat Ghassul (Mallon *et al.* 1934: 67-70; Hennessey 1969: fig. 12:1), Abu Matar (Perrot 1955: fig. 18:b), Horvat Beter (Dothan 1958: fig. 11:20), Neve Ur (Perrot *et al.* 1967: fig. 13:1, 2), and sites on the Golan Heights (Epstein 1998: pl. XXXV).

Basalt and limestone mortars (Fig. A5:5, 6) are not unique to the Chalcolithic, but have been reported from many sites of this period – Horvat Beter (Dothan 1958: fig. 11:1, 5, 6), Neve Ur (Perrot *et al.* 1967: fig. 14:1), Teleilat Ghassul (Hennessey 1969: fig. 12:4), Abu Matar (Perrot 1955: fig. 18:b), Grar (Gilead 1995: fig. 3.7:1-3), sites on the Golan Heights (Epstein 1998: pl. XXXVI), and others.

The small shallow bowls, prepared on unworked pebbles (Fig. A5:7), were present in assemblages from many sites – Horvat Beter (Dothan 1958: fig. 19:24), Grar (Gilead 1995: 325), and sites on the Golan Heights (Epstein 1998: pl. XXXVI: 8, 9). At Abu Matar such a bowl, with ochre residues, was found (Perrot 1955: 79).

The quantity of pestles at our site (Fig. A6:1-3), even relative to other excavated sites, was high. Furthermore, according to information from workers who worked at the site, some time ago more pestles were taken from the site. In the detailed report from Grar (Gilead 1995) these tools are missing. From the Golan Heights a single piece was reported (Epstein 1998: pl. XLI: 25). About one-third of the pestles in our assemblage are of the elongated conical type. Similar tools (of smaller dimensions) were reported only from Teleilat Ghassul (Mallon *et al.* 1934: pl. 38:9, 10). Information about roughly cylindrical barrel-shaped pestles and pounding pestles with rounded ends is scarce.

Most publications from excavated sites on grinding stones supply only general information. The axe-shaped grinding stone (Fig. A6:4) was not mentioned in the publications which we checked. The match in shape to the conical cup-marks (Fig. A5:1a) proves that these were not cultic axes or burial offerings, but grinding tools.

Round grinding stones, similar to those from our assemblage (Fig. A6:5), were published from the Golan Heights (Epstein 1998: pl. XLI). Small grinding stones of various shapes (Fig. A6:6, group c. above), have been defined as 'handstones' in many publications. This definition could be misleading, as flint tools named 'biface', are also called 'handstone', but their purpose is completely different. In some cases these small grinding stones were included in the category of 'varia' (Perrot *et al.* 1967: 217; fig. 13:11-13).

At this stage the materials processed with the grinding tools are not known. The possibilities include food, minerals, and paint processing. Possibly some of them also served for leather tanning (Adams 1988: 307).

Point-sharpening tools (Fig. A6:7) were known from the Neolithic. They served, apparently, for sharpening points of perishable materials such as wood or bone. Parallels for these were found at Teleilat Ghassul (Mallon *et al.* 1934: pl. 38:6; Hennessey 1969: pl. 15b).

The round polished basalt fragment with a conical hole is of special interest. At first it was assumed that this was a weight for a digging stick, but a close investigation showed that its circumference was carefully worked and polished to a disc. The circular conical hole was polished by a continuous rotational motion. A tool, similar in shape and dimensions, was published from Metzger (Dothan 1959: 28), and defined as a potter's wheel. Dothan relied on the publications of Megiddo XVIII and Beth Yerah II (Dothan 1959: 28). A fragment of a similar tool was found at Horvat Beter (Dothan 1958: 27). Possibly these tools were indeed potters' wheels. In this case the burnishing tools mentioned above would have served for

the production of pottery vessels (Amiran 1971: 22-23).

A parallel for the short cylindrical basalt artifact with a round depression on one end (Fig. A6:8) was found only on the Golan (Epstein 1998: pl. XLIV:5).

Lower grinding slabs (Fig. A6:9) were found at many contemporary sites: Horvat Beter (Dothan 1958: 27, fig. 19:9), Neve Ur (Perrot *et al.* 1967: fig. 14:6), Grar (Gilead 1995: fig. 4.7:3-5), the Golan Heights (Epstein 1998: pl. XXXIX:2), and many more.

SUMMARY AND CONCLUSIONS

In contrast to assemblages at excavated sites, surface finds from a survey present only primary and partial data, and not the complete picture of the site. The character of our random sample of artefacts was influenced by the diagnostic capability of the surveyors, geological processes, erosion, and other factors. All these make exact chronological conclusions quite difficult, and the value of statistical comparisons limited.

The flint tool assemblage gives evidence of a first occupational phase, at this stage of a still undefined scope, during the Pre-Pottery Neolithic B. During the long duration, about 1500 years, of the Chalcolithic period, the material culture changed and a differentiation between groups – clusters – occurred: Teleilat Ghassul, Beer Sheva, Besor, and others (Gilead 1995: 473-476). According to the flint, basalt, and stone assemblages, and the rock-cut installations, after a time gap, at some stage during the Chalcolithic, a second and larger occupational phase took place at the site. At this stage it is still impossible to define the duration of this occupation and its position in the chronological framework. Furthermore, it is difficult to define whether the character of the flint and stone assemblages point to regional traits or to a specific specialization.

The evidence for other occupational phases is insubstantial. No clear finds from the Pottery Neolithic period, the Wadi Rabah phase, or the Bronze Age were found.

For a confirmation of our preliminary assumptions and for a wider range of conclusions about the character, the chronology and the cultural entities of the site, further research and analyses are essential. A future excavation could contribute to the knowledge about the above-mentioned periods in a region which to date has been barely studied.

The research of use-wear of flint and stone tools, including microscopic analyses and experimentation with replicas, would contribute further insights about the functions of the tools and the economy.

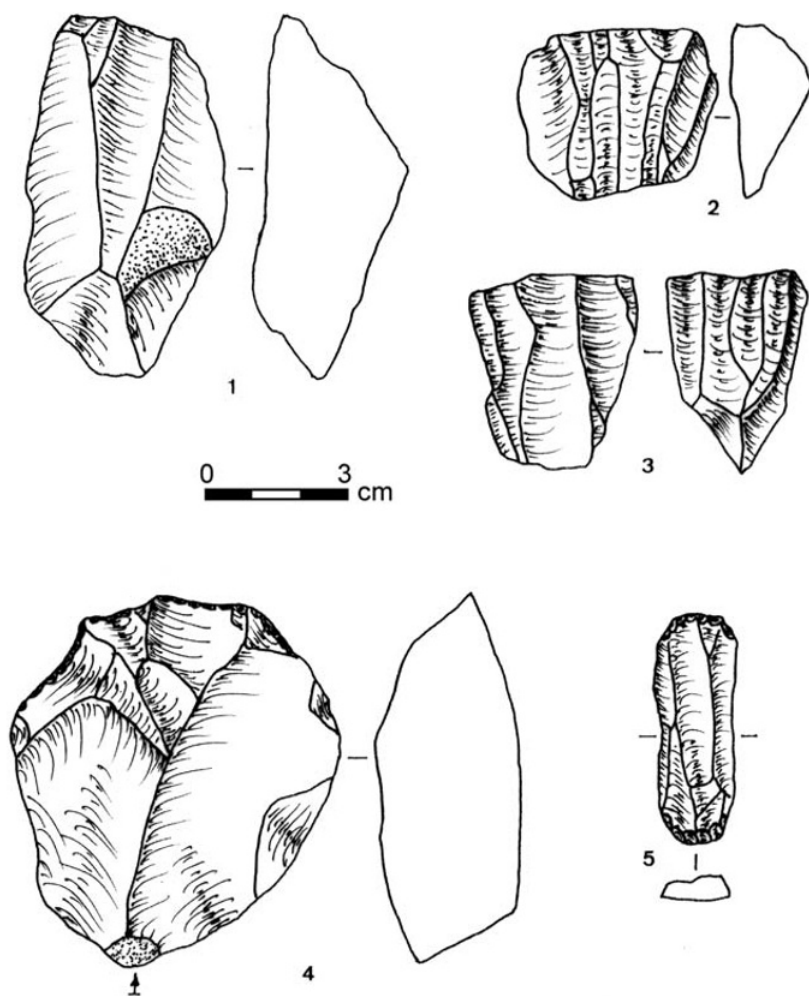


Fig. A1: 1. Naviform core; 2. Bladelet core; 3. Pyramidal core; 4. Coarse scraper; 5. Double scraper.

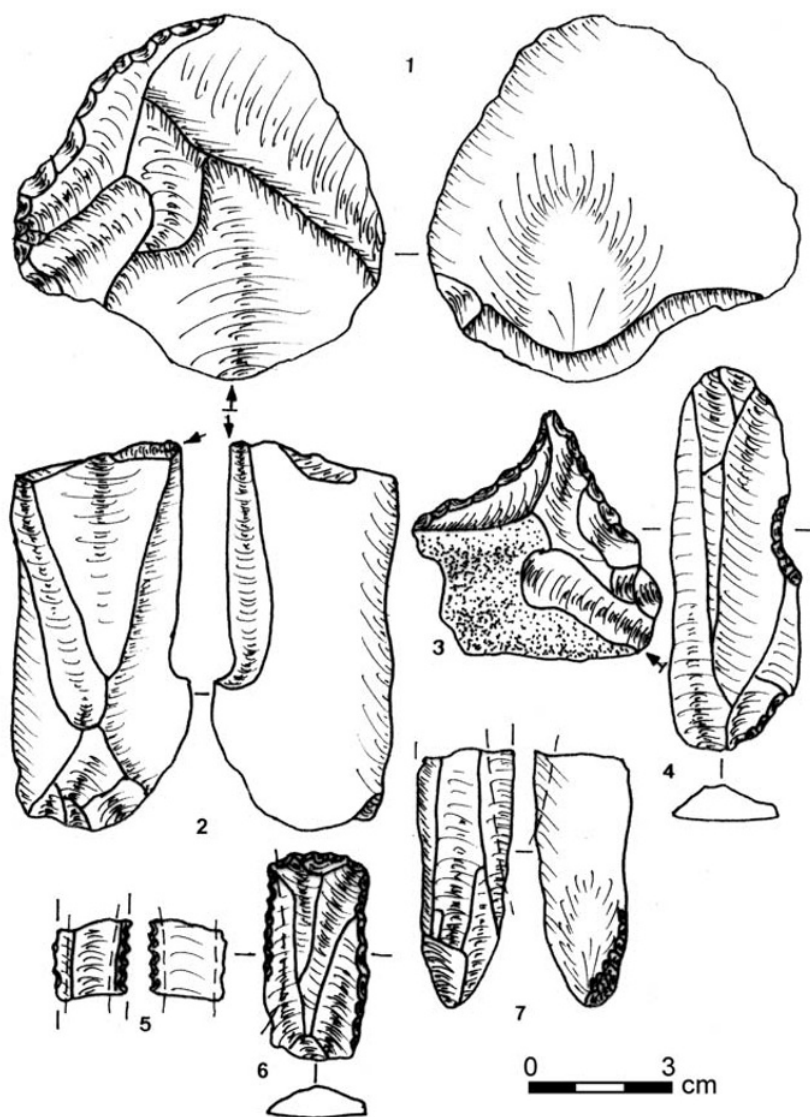


Fig. A2: 1. Dejeté scraper; 2. Burin; 3. Awl; 4. Notch; 5, 6. Sickles; 7. Reaping knife.

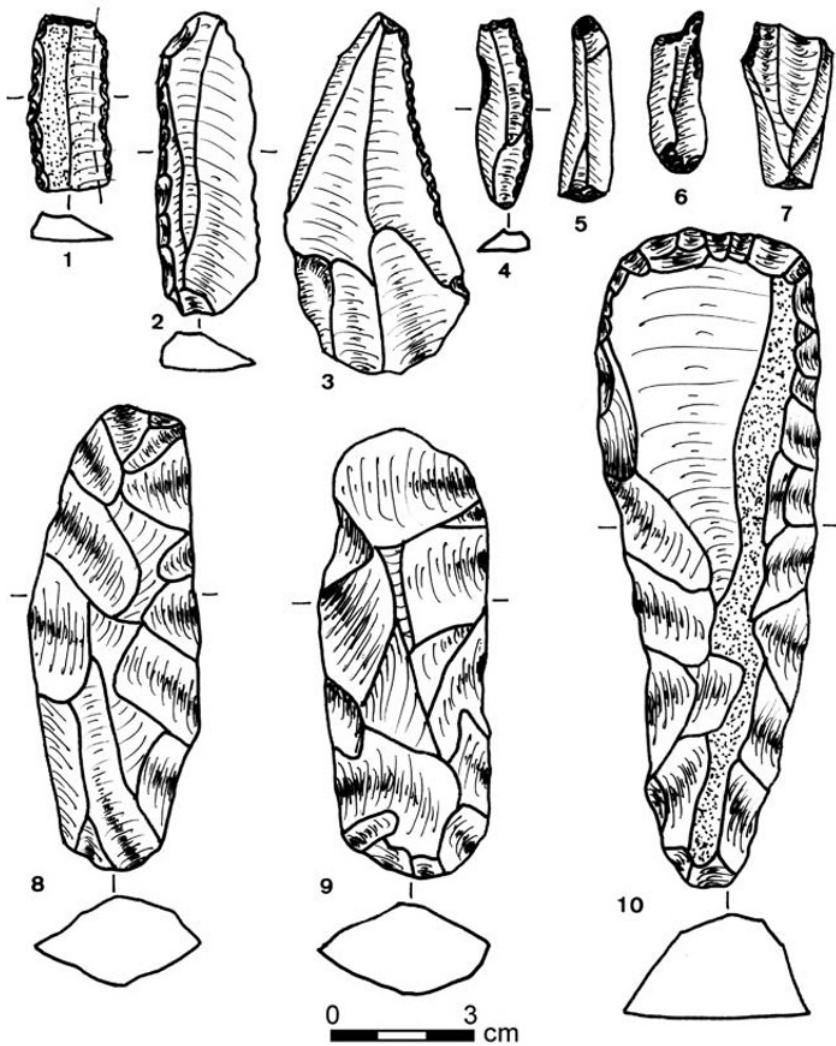


Fig. A3: 1. Sickie blade; 2. Backed knife; 3. Retouched blade; 4. Backed bladelet; 5. Micro end-scraper; 6. Awl on bladelet; 7. Microburin, 8, 9. Axes; 10. Adze.

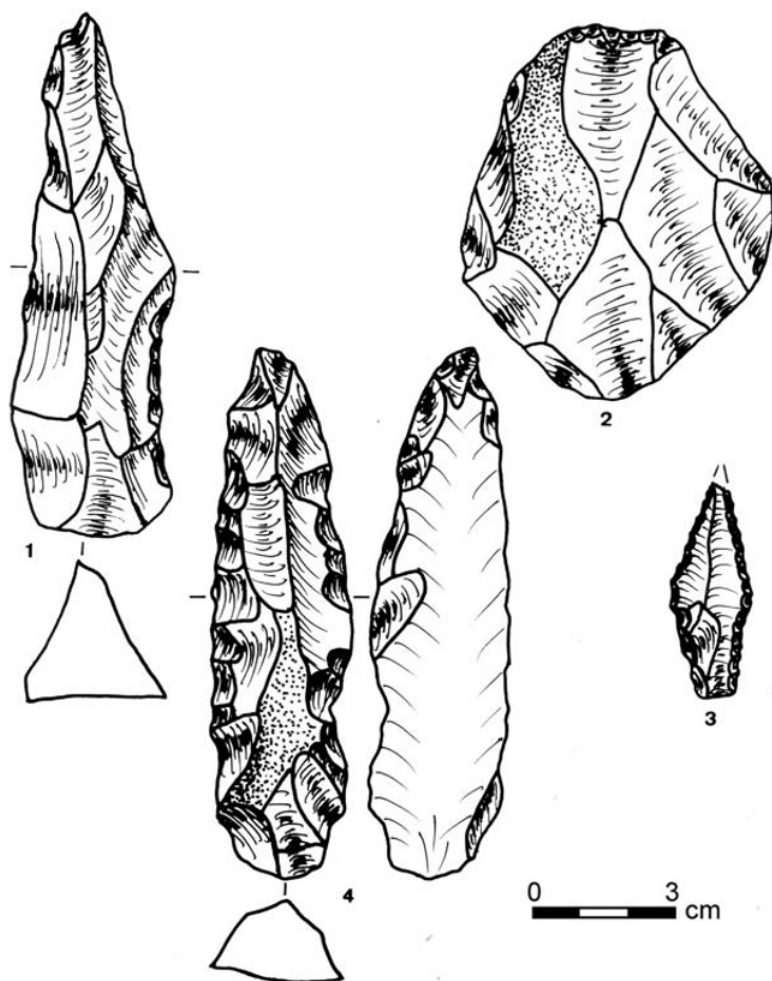


Fig. A4: 1, 4. Picks; 2. Chopper; 3. Byblos arrowhead.

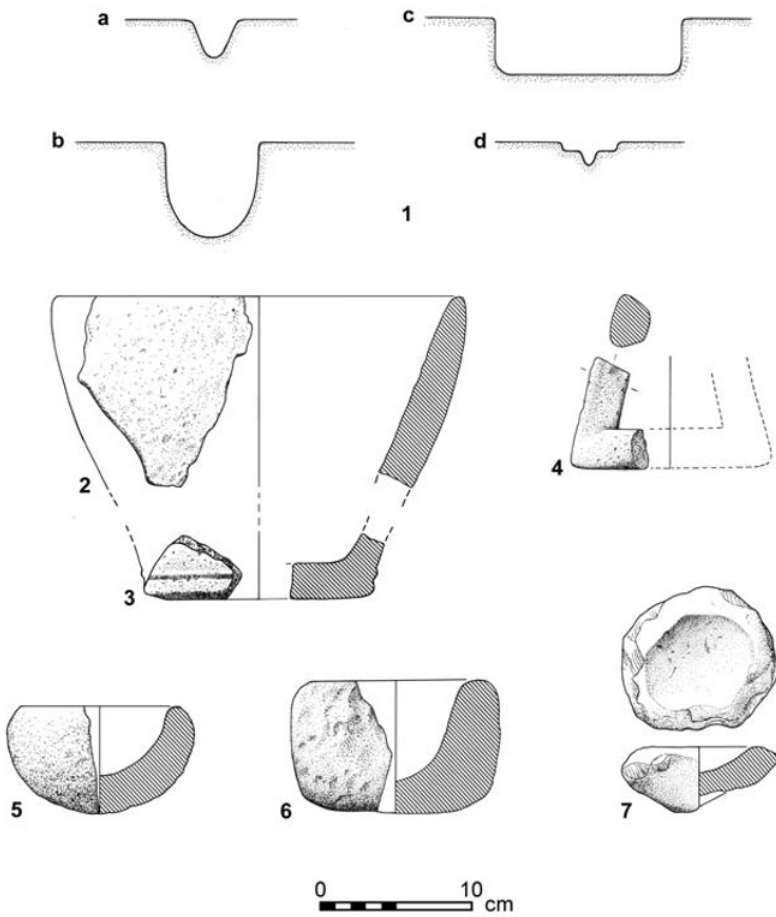


Fig. A5: 1. Cross-sections of rock-cut installations; 2. Rim of conical bowl; 3. Base of conical bowl; 4. Base of pedestal; 5. Globular mortar; 6. Mortar with flat base; 7. Small bowl on pebble.

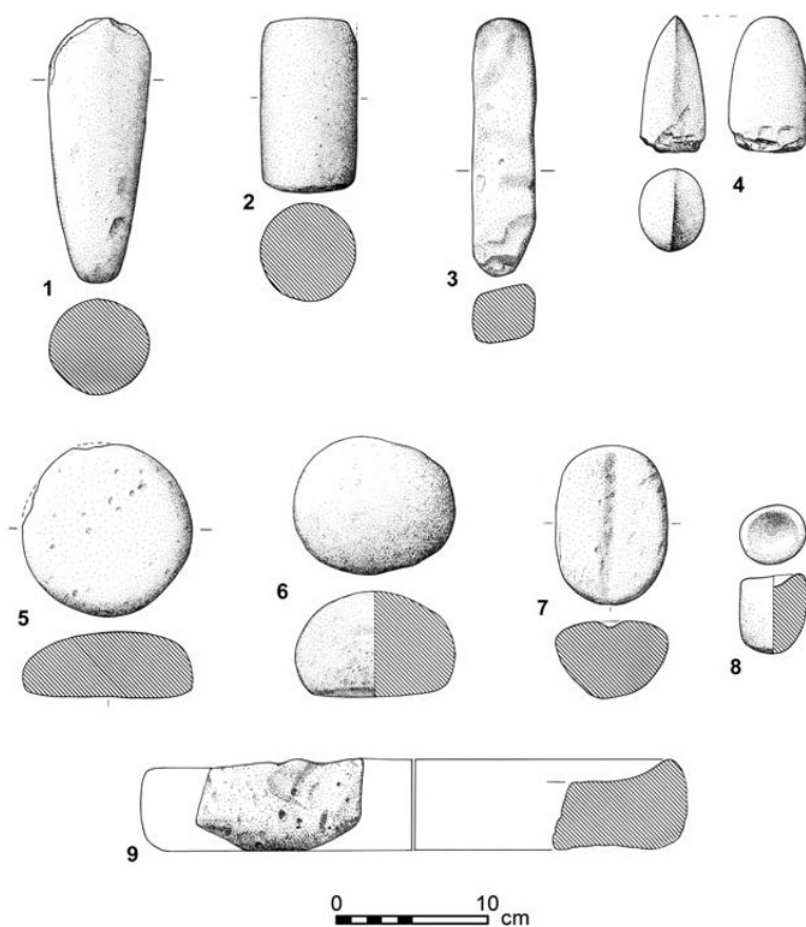


Fig. A6: 1. Conical pestle; 2. Cylindrical pestle; 3. Pounding pestle; 4. Axe-like grinding stone; 5. Discoid grinding stone; 6. Loaf-shaped grinding stone; 7. Point sharpener; 8. Cylindrical basalt artifact with a round depression; 9. Lower grinding slab.

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APPENDIX B

THE 'ARUNAH PASS

Adam Zertal

INTRODUCTION

The Roman 'Via Maris' ('way of the sea'), the ancient road connecting Egypt and Mesopotamia, crossed Sinai and then went northwards along the hills of Samaria. After passing through a hilly obstacle discussed below, it entered the Valley of Jezreel and from there, via different roads, to Damascus and the Euphrates (Abel 1938: 218-219; Meshel 1967; Aharoni 1979: 45-54; Rainey 1983; Zertal 2004: 31).

The term 'way of the sea' is mentioned just once in the Hebrew Bible, in an extremely difficult text (Isaiah 8:23). The combination there "the way of the sea, the land beyond the Jordan, Galilee of the nations ('goyim')"

has no parallels, nor has its meaning ever been agreed upon (for interpretations see Alt 1953; Emerton 1969; Hogenhavn 1987; Sweeney 1994; Childs 2001: 78-81; Zertal 2010: 250-251). Moreover, the connection of Isaiah's 'way of the sea' to the Roman 'Via Maris' is not at all automatic. Another 'way of the sea' is mentioned in I Kings 18:43, in the story of Elijah. This 'way of the sea' could possibly be the road which crosses the southern Carmel to the east (Artzy 1998), again having no connection to our issue.

In general, a 'road' in the biblical narratives and/or the Egyptian texts is rarely built or paved. Before the road-making in the Roman period no well-established and maintained roads existed, and 'ways' or 'roads' were no more than dirt paths, similar to those in use to this very day in the hill country or the Negev desert.

To identify an ancient 'road' therefore, we should use the historical documents, together with a thorough knowledge of the topography, water sources, and the original vegetation of the area.

THE THREE PASSES

The topography of ancient Canaan is not extreme by elevation, cliffs, etc. The obstacles to movement in the Mediterranean regions were rivers, marshes, and swamps in the plains, and thick vegetation in the hill country. In our case, the Sharon Plain is separated from the Jezreel Valley by the

hills stretching between the Carmel and Samaria.

The first and foremost relevant narratives concerned are those of Thutmosis III's first campaign, the "Satirical letter" of Hori from the days of Ramses II (ANET: 475-480), and the campaign of Shishak of the 22nd Dynasty (ANET: 263-265; Simons 1937: 89-101). While Thutmosis speaks of the three passes, the two latter relate only to the 'Arunah Pass.

The roads discussed in the Yehem council of Thutmosis are these: firstly, the southernmost ('eastern') which goes to Ta'anach; secondly, the northern one heading to Djefti; and thirdly, the central one, that of 'Arunah (Fig. B1; ANET: 235; Urk. IV: 650.8-12; Mizrachy 2005: 425).

The generations-long consensus that the 'Arunah Pass was the most important of the three was only based upon its relatively detailed description. However, following our archaeological survey along the pass, and the new understandings which followed, this road emerges as difficult and dangerous and the least-used of the three. This issue is discussed at the end of this Appendix.

The main theme of Thutmosis story, typical of other royal Egyptian documents, was called *Königs-nouvelle*, a term which means glorifying the courage of the King (Spalinger 1974; 1979; Hoffman 2004). It should be treated, therefore, with general criticism and scepticism. However, it seems that the Egyptian text has in this case provided us with a fair amount of topographic detail, together with the psychological value of the pass.

The story of Thutmosis opens at Yehem, where the council of the King's generals took place. The Canaanite army, headed by the King of Kadesh, was waiting for the Egyptians near Megiddo, and the council discussed the roads to the battle site. Alt has identified Yehem as Tel Assawir (Tel Esur – Site 30), at the western entrance of the 'Arunah Pass (Alt 1914; 1932), while Albright has correctly shown that the site should be south of Jatt (Gath-Padalla – Albright 1923). He based his conclusion on the fact that the council speaks of all three roads as starting north of Yehem, a condition not suiting Tel Esur/Assawir (Fig. B1).

Finally, Yehem was correctly identified at Khirbet Yemmeh, south of Jatt. The small fortified tell dating to the Middle and Late Bronze Ages was sounded by Garstang (1931: 85, 90, and pl. IX), and the identification is also supported by the similarity between the ancient and modern names.

The Canaanite coalition, according to the Annals, was divided into two forces, with Megiddo between them: the first headed south near Ta'anach, and the second was waiting north of Megiddo. The text, translated by Wilson, is as follows:

“They said in the presence of his majesty: What is it like to go [on] this [road] which becomes (so) narrow? ...Will not horse (have to go) after [horse, and the army] and the people similarly? Will the vanguard of us be fighting while the [rear guard] is waiting here in 'Arunah, unable to fight? Now two (other) roads are here. One of the roads – Behold, it is [to the east of us], so that it comes out at Ta'anach. The other – Behold, it is to the north side of Djefti, and we will come out to the north of Megiddo. Let our victorious Lord proceed on the one of [them] which is [satisfactory to] his heart, (but) do not make us go on that difficult road!” (ANET: 235).

The three passes are identified by sites. The eastern comes out at Ta'anach; the central is that of 'Arunah; and the northern site is marked by Djefti.

THE CENTRAL ROAD – THE 'ARUNAH PASS

Previous scholarship concerning this road has been based purely upon textual analysis (Nelson 1913; Faulkner 1942; Noth 1943; Helck 1971; 1972; Spalinger 1979; Burkhardt *et al.* 1984; Goedicke 2000; Redford 2003). The area south of the pass (today's Wadi 'Arah) had never been surveyed, while that in the north is better known, especially in the last few years (Olami 1981: 15-22; Gadot 2006). The geology of the area was explored by Sass (1968), but just two works (Vilnay 1960; Golani 1966) dealing with the pass and its history were published prior to the publication in Hebrew of this Volume, with the new understandings gained concerning the pass.

The 'Arunah Pass is a geological fault between the Umm el-Fahm anticline and the soft Senonian hills to its north (Fig. B2). The pass is 20 km long, and is divided into three sections, from west to east: the 6 km-long, wide 'entrance' from Tel Esur to 'Arah ('Arunah); the 'Arunah Valley, 3 km-long, with springs; and the relatively narrow 11 km-long section from Khirbet Rabzeh to the junction near Tel Megiddo. The three tells along the pass are, from west to east, Assawir, 'Arunah, and Megiddo. The western entrance to the road has always been marked near Tel Esur (Mazar 1975: 149 – map 149; Aharoni 1979: 50, and map 3; Mizrachi 2005: map 4). Its presence raises the problem why this tell, equivalent to Megiddo by its position, did not continue its existence and develop from the Bronze Age period onwards (see Appendix F and Zertal 2003). Moreover, the tell is neither identified nor mentioned in known texts (In 1977 I raised the possibility that this can be Djefti of the Thutmosis Annals; This suggestion should probably be renewed now, in light of the renewed excavations; cf. Na'aman 1984: n. 26.).

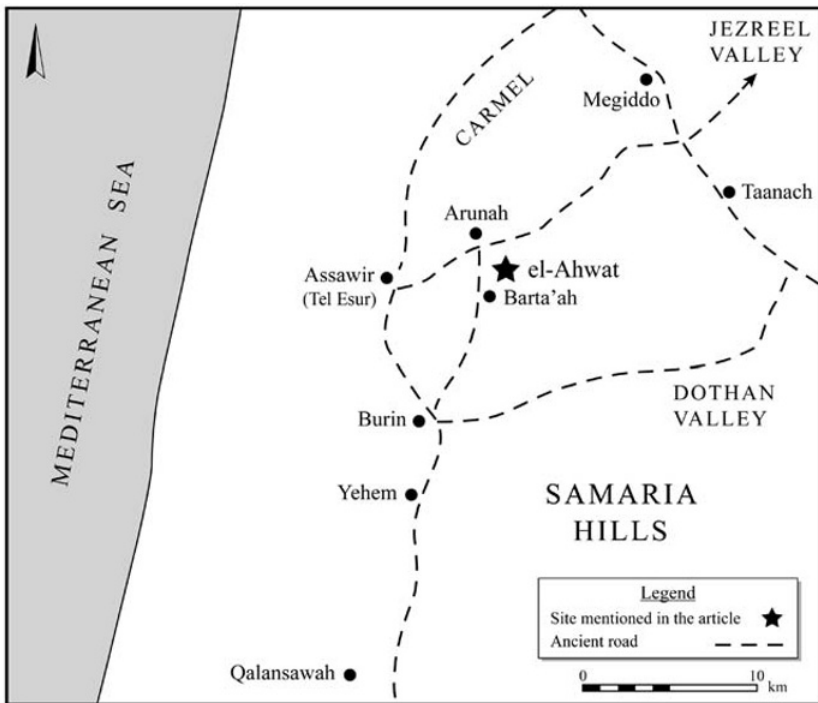


Fig. B1. The 'way of the sea' and its branches, with the sites discussed.

A possible reason for its inferiority to Megiddo may be that the main entrance into the pass was not at Tel Esur, but somewhere else. This option, explored during the survey, was found to be a shorter and more convenient road. Its existence also opens new directions for the understanding of the Egyptian texts.

The geographical problem with the Tel Esur entrance is its being a marshy area: during the autumn, winter and spring the territory near the tell was regularly flooded (Uchmani 1967: 14; Jibly 2004: 61), and movement was difficult or impossible. It is not a mistake, therefore, that in the *Annals* the next station moving northwards from Yehem (or Burin) was 'Arunah (ANET: 235; Mizrachi 2005: 424), and also in Shishak's list (Simons 1937: nos. 33-32), with no site in between.

The 'new' branch explored in the survey begins at modern Baqa-el-Gharbiyeh (ancient Burin), entering Wadi Samantar, and then straight to 'Arunah (Fig. B2). It is 2–3 km shorter than the Tel Esur one, and by-passed its marshes. Three Iron Age sites are located along this road (Sites 94-95, 104), together with a square Roman road fortress (Site 37).

The Samantar road is relatively narrow, located inside thick vegetation and old oak trees. This is beautifully described by a 19th century text of a Jerusalemite Jew, who travelled along this road from Jerusalem to Umm el-Faḥm in 1878 in search of citrons: "There in Qalansawa we rested till afternoon, and then went to the garden of Barta'ah. This road goes through a thick forest of large oaks and pistacia trees, with long branches...this is an awful country...a narrow and twisting path that makes you turn your horse to the right and the left, and sometimes turn backwards, meeting oak trees that block your path, so hard that we could not believe that ever people moved here...when the sun came down we finally saw the garden of Barta'ah" (Krassel 1955: 110-113).

Interestingly enough, this text, written by Joel M. Solomon, a leader in 19th century Jerusalem, is very similar to that of Thutmosis III's generals. The pass is marked in the British Palestine Exploration Fund survey as a good track (Conder and Kitchener 1879: sheet VIII, JI-Kk).

The same pass was used in modern times (the 30s of the 20th century). Meiron and Kabha (1993: 119-120), having collected traditions about Wadi 'Arah, write: "Wadi al-Muraba'at (=Samantar), between Baqa and Barta'ah, was a short way from the north of the country to Tulkarm. Some robbers used to attack the trade caravans there".

The Thutmosis III Annals raise another problem. The council of the King took place in Yehem, but the generals use the term 'here in 'Arunah' (ANET: 235; Mizrachy 2005: 425) which contradicts the location at Yehem. Nelson (1913) saw the inconsistency, but did not explain it. His idea that it was a simple mistake is not reasonable. We suggest that the description of the narrow road relates to Wadi Samantar, and the two parts of the text have been mixed. The council indeed began in Yehem, and then the army moved through the pass of Samantar; so they describe what they saw in the immediate past, probably to warn the King from continuing (The mixture of events that happened at different times is very common in Egyptian art; cf. the camp of Ramses II in the battle of Kadesh [Yadin 1963: pl. on 156] where events at different times occur together). This notion becomes clearer when comparing two texts of the 'Arunah Pass – of Thutmosis and Hori (Papyrus Anastasi I). While the first speaks of a narrow path only, the latter describes steep slopes and thick forest on both sides of the pass, together with other frightful dangers.

The text of Hori speaks as follows: "Behold, the ambushade is in a ravine two thousand cubits deep, filled with boulders and pebbles. Thou makest a detour, as thou graspest the bow. Thou makest a feint to thy left, that thou

mightest make the chiefs to see (but) their eyes are good and thy hand falters. "Abata kama ir, mahir ne'am!" (Thus) thou makest a name for every mahir, officers of Egypt! Thy name become like (that of) Qazardi, the Chief of Aser, when the bear found him in the balsam tree. The narrow valley is dangerous with Bedouin, hidden under the bushes. Some of them are four or five cubits (from) their noses to the heel, and fierce of face. Their hearts are not mild, and they do not listen to wheedling. Thou art alone; there is no messenger with thee, no army host behind thee" (ANET: 477).

The differences between the two descriptions are clear. While Thutmosis does not mention high slopes at all, something one could surely expect, Hori concentrates on them; in Thutmosis' time no danger of Bedouins, or

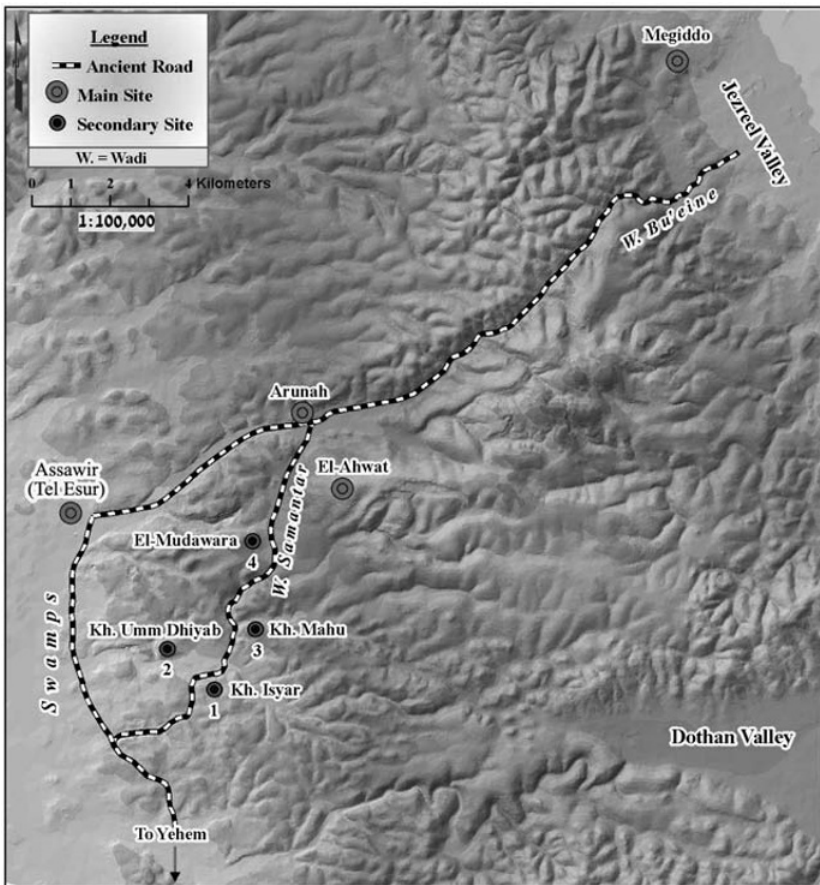


Fig. Bz. Map of the 'Arunah Pass. Numbers of sites are according to the text.

other tribes or names of enemies, is mentioned: Hori, on the contrary, is very afraid of groups such as the children of Aser with their leader Qazardi (Avi-Yonah [1973: 783] discusses the possibility that the name 'Qazardi' is West Semitic in origin). It can be concluded therefore, that the two texts speak about different places and different situations.

Yet what is the reality behind each of the narratives, 200 years apart?

Firstly, the path of Samantar has no cliffs or slopes, which well suits the Thutmosis text between Yehem and 'Arunah; secondly, in the 15th century BCE the pass is completely unsettled, as also emerges from the survey. The 'foe' of Thutmosis is waiting only outside the pass toward Megiddo (Mizrachy 2005: 425), with no clue as to the dangers along it. The generals of Thutmosis complain mainly about the problem of not using the whole Egyptian force, because of the narrowness of the road.

In the second case, the letter of Hori, the reality is sharply different. The topography of the pass is narrow with high sides, and the area is full of hostile populaces or tribes, who lie in wait for those who attempt the passage. Only the name – Megiddo or 'Arunah Pass – and the thick vegetation are common to both.

In addition, the Hori letter must relate to a different section of the pass, with high and narrow topography. Since no other part of the pass suiting this description has been mentioned, nor do they exist along the road, we suggest a new candidate, discovered during the survey. This is the 2–3 km long, deep Wadi Bu'eineh, which continues Wadi 'Arah to the east, and exits to the plains of Jezreel near modern Zalafeh. It is a narrow wadi, with steep sides with thick vegetation, exactly matching Hori's description.

Historically and ethnically speaking, the area in the 13th century BCE was completely changed. A concentration of 10–12 Iron Age I sites have been surveyed in the vicinity of the pass. It was populated not only by these 'new' sites but, above all, the town of el-Ahwat (Site 28) had been founded. This town, discovered in 1992 and excavated from 1993 to 2000 (Zertal 2008; 2010; 2012), is a heavily fortified settlement, 3 ha in area. It was a short-lived site, dating approximately between 1220 and 1160 BCE. The city was divided by inner walls into 'quarters' A–E, with a city gate and a 'governor's house'. The rich collection of small finds includes Egyptian and Egyptianized scarabs, with one bearing the royal name of Ramses III; various seals; an ibex ivory head, part of an elegant vessel; a head of a bronze linchpin of an Egyptian (?) chariot, etc.

The site of el-Ahwat is exceptional from several aspects: it is the only fortified city founded on virgin soil during the Iron Age I in the southern

Levant; it has architectural elements alien to Canaan of the period: corridors in the city wall, round stone-built huts, a furnace for forging iron, etc. This combination led the excavators to suggest a foreign origin for the settlers, probably Nuraghic Sardinia. If so, it was the Sherden (*Šrdn*) of the Egyptian texts who may have been settling the site, presumably as a fort under Egyptian rule in the beginning (on these 'forts' *cf.* ANET: 262 and Gardiner 1947: 293; for the Shardana *cf.* Loretz 1995; for el-Ahwat and its interpretations see Ugas 2008; Zertal 2008; 2010; 2012).

Back to the 'Bedouin' of Hori: it has long been suggested that the name 'Aser' may relate to families of the Israelite tribe Asher, whose people were most scattered among the tribes (ANET: 477: n. 42; Simons 1937: 162; Gardiner 1947; Avi-Yonah 1973: 783-784; Aharoni 1979: 51). It can be surmised now that those sites in the vicinity of the 'Arunah Pass, a basically Manassite territory, also included Asherite families. The mercenaries/robbers of el-Ahwat may be easily added to these gangs.

THE CAMPAIGN OF SHISHAK

Our interest in Shishak's list concerns the order of the sites along the pass of 'Arunah. This order is not at all agreed upon, which has resulted in different views on how to arrange the sites (Simons 1937: 89-101; Mazar 1974: 236; Aharoni 1979: 323-327). Here again, as in Thutmosis' list, there are no stations on the south-north road between Burin (Baqa el-Gharbiyeh – Site 122) and 'Arunah (Site 13). On the other hand, between Megiddo and 'Arunah there are no less than four stations in the list: (28) *adr* (Adar); (29) *id-hmlk* (Yad-Hamelech?); (30) --- (31) *hnm* (Honim?) (Mazar 1974: 234-242; Aharoni 1979: 325).

The problem is this: from Megiddo to 'Arunah not one Iron Age candidate for these stations is known (Mazar 1974: 239), whereas along the 13 km-long Samantar road there are three good sites with early Iron Age II pottery of the 10th century BCE. These are, from south-west to north-east: Khirbet 'Isyar A (Site 104); Khirbet Umm edh-Dhiyab (Site 94); and Khirbet Mahu (Site 95 – Fig. B2). All three, located upon, or near, the road, have Iron Age I and II sherds. There are also some clues for possible name preservation. Ader or Adar (no. 28 of Shishak) may have been preserved in the name 'el-Medwar' near Khirbet Mahu (Conder and Kitchener 1879: sheet VIII, Kl.; no. 4 in Fig. B2). Although the name is Arabic (meaning 'the round'), it also fits Kampffmeyer's rules of name preservation (Aharoni 1979: 111-118). The other, *hnm* or Honim (no. 31 of Shishak), may have been

preserved in the strange place name Mahwah (today Mahu). In this case, the 'n' has been deleted, and the order of the consonants changed in the rest of the name (*hnm* = *hm* = *mh*). Be these new identifications correct or not, the archaeological reality better suits a different order of Shishak's list-reading.

WHAT IS THE SIGNIFICANCE OF THE PASS?

In a previous paper I discussed the psychological significance of the 'Arunah Pass (Zertal 2000). Against the 'classic' opinion of its importance, we have suggested that the pass was above all a 'test' for the courage of the kings (or others). For trade and regular use, it seems that the caravans used the southern pass, that of the Dothan Valley, and not the dangerous and narrow 'Arunah road.

To understand this idea, the answer of Thutmosis to his generals emphasizes his courage (ANET: 235: 40-50). He shows that, in spite of the difficulties, he will take the dangerous road. Wilson, in his introduction to Hori's letter, writes (ANET: 475): "Hori responded in a lofty and sarcastic vein, attempting to expose the weaknesses in his correspondent's qualifications for office." He did it by various ways, one of which was to show the courage needed for the 'Arunah road.

Secondly, had this pass been in regular use, it should have been mentioned in most or all the New Kingdom diaries and other sources. And yet, Amenhotep II, for instance, among others, does not use it (Na'aman 1984).

Moreover, this pass is not among the many roads mentioned in the Hebrew Bible. It was only during the Roman Period that a road was established and paved, when Hadrian built the camp of the 6th Legion at Legio (Avi-Yonah 1976: 74-75), near Megiddo, and connected it to the capital city of Caesarea Maritima on the Mediterranean coast.

Based upon several historical sources, it can be suggested that the southern road, which goes into the Jezreel Valley through Jenin or Wadi Hasan, was used more regularly than the 'Arunah one (Genesis 37:25; *War II*: 284, 510; Conder and Kitchen 1882: 50; Albright 1923: 10; Zertal 1990). In the Medieval and Mamluk periods, the Jenin-Qaqun road became most important, especially as the mail-road that led from Damascus to Cairo, and was also used in the Ottoman period (Volney 1798-1799: 33-34; Tschelebi 1980: 34; Sezgin 1994: 33-38). In all these periods, no mention is made of the 'Arunah Pass.

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APPENDIX C

COIN FINDS FROM VOLUME THREE OF THE MANASSEH HILL COUNTRY SURVEY

Shay Bar

A catalogued description of the coin surface finds is presented here as an Appendix with a period of 1700 years embodied in the numismatic finds, from the Hellenistic period to the Middle Ages. The Appendix presents the origins and dates of minting of the coins: some of them minted locally, but most minted throughout the empires that controlled the land in these periods.

Within each site the coins are numbered chronologically. The information about each coin is in the following order:

- A. General description: a. Period; b. Ruler and regnal years; c. Material; d. Diameter; e. Weight.
- B. Description of obverse and reverse sides, including inscriptions, date and mint.
- C. References.

Site No. 125:

Khirbet Shamsin

- 1. Hellenistic, Alexander Yannai (103-76 BC), Bronze, 12.4-13 mm, 1.42 g.

Obverse: worn.

Reverse: Cornucopia, worn inscription and pearls around edge.

Reference: Kadman 1957: 28.

Site No. 131:

Khirbet Samarra

- 1. Late Roman, Constantine II (337-340 CE), Bronze, 15.5 mm, 2.65 g.

Obverse: Ruler facing right, adorned with diadem, armoured.

Around: worn Latin inscription and pearls around edge.

Reverse: worn.

Reference: Sear 1981: no. 3910.

2. Late Roman, Arcadius (393-408 CE), Bronze, 13 mm, 1.09 g.

Obverse: Ruler facing right, with diadem, armoured, pearls around edge.

Reverse: Victoria dragging captive to left, pearls around edge.

Site No. 143:

Khirbet el-Qusur

1. Byzantine, Justinus I (518-527 CE), Bronze, 30 mm, 13.76 g, Follis-40 Nummia.

Obverse: Worn. Ruler facing right, armoured.

Around: worn Latin inscription '...VSTI... N..P..'

Reverse: Large 'M', below: 'CON', Constantinople mint.

2. Medieval, Bronze, 18.8-20.8 mm, 2.40 g, Fils.

Obverse: worn Arabic inscription.

Reverse: worn Arabic inscription.

3. Medieval, Bronze, 15.7-18 mm, 1.40 g, Fils.

Obverse: worn Arabic inscription.

Reverse: worn Arabic inscription.

Site No. 156:

Khirbet Barquneh

1. Byzantine, Justinus (527-565 CE), Bronze, 28.5-30.5 mm, 14.45 g, Follis-40 Nummia.

Obverse: Worn. Ruler facing right, adorned with diadem, armoured.

Around: worn Latin inscription '...VSTINI... AN...', Ring rope decoration.

Reverse: Large 'M', cross above it and at left, six-pointed star at right.

Below: 'CON', Constantinople mint.

Reference: Sear 1974: no. 158.



Fig. C1: Reverse of coin 1 from Khirbet Shamsin.



Fig. C2: Obverse of coin 1 from Khirbet Samarra.

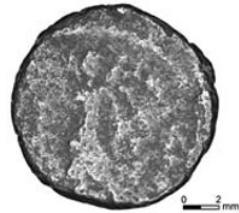


Fig. C3: Reverse of coin 2 from Khirbet Samarra.



Fig. C4: Reverse of coin 3 from Khirbet el-Qusur.

Site No. 158:

Jami' Tubrus

1. Medieval, Mameluk, en-Nasr Aladdin Muhammed? (1310-1341 CE), Bronze, 18 mm, 2.88 g, Fils.

Obverse: worn Arabic inscription above and below horizontal line.

Reverse: worn Arabic inscription, pearls around edge.

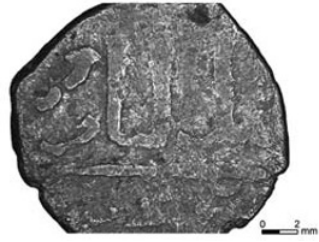


Fig. C5: Obverse of coin 1 from Jami' Tubrus.

Site No. 195:

Khirbet Bajjurah

1. Roman, Valerian (253-260 CE), Bronze, 17.5-19.5 mm, 2.86 g, Antoninianus.

Obverse: Ruler facing right, adorned with a crown of rays, clothed, armoured.

Around: Latin inscription: 'PCPLICVALERIANUS...VG'.

Reverse: Figure on right (Valerian?) with spear in right hand facing left, receiving a dagger? from a female figure on left facing right.

Around: Latin inscription: 'RESTI...EN...'.
Reference: Sear 1981: no. 2796.

2. Late Roman, Constantine II (337-340 CE), Bronze, 16.5 mm, 1.86 g.

Obverse: Ruler facing right, adorned with diadem, armoured.

Latin inscription: '...ONSTANTINUS...', pearls around edge.

Reverse: Two soldiers standing with two banners between them, spear in outside hands, inside hands on shield.

Around: worn Latin inscription.

Reference: Sear 1981: no. 3851.



Fig. C6: Coin 2 from Khirbet Bajjurah.

3. Late Roman, Constantine II (337-340 CE), Bronze, 15.5 mm, 1.66 g.

Obverse: Ruler facing right, adorned with diadem, armoured.

Latin inscription: 'CONSTAN...TINUSAVG', pearls around edge.

Reverse: Two soldiers standing with a banner between them, spear in outside hands, inside hands on shield.

Latin inscription: '...IAEXERC..TUS', pearls around edge.

Reference: Sear 1981: no. 3855.

4. Byzantine, Tiberius (582-602 CE), Bronze, 26-30 mm, 10.13 g, Follis-40 Nummia.

Obverse: Ruler facing right, adorned with diadem, clothed, sceptre in left hand.

Around: worn Latin inscription '...HU...TIMIK...!.

Reverse: Large 'M', cross above it, ANNO (year) at left, U (5) at right.

Below: 'THEUP', Antioch mint.

Reference: Sear 1974: no. 532.

5. Early Moslem, Bronze, 17.5 mm, 2.48 g, Fils.

Obverse: worn Arabic inscription, pearls around edge.

Reverse: worn Arabic inscription inside two concentric circles of pearls around edge.

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Fig. C7: Coin 3 from Khirbet Bajjurah.



Fig. C8: Reverse of coin 5 from Khirbet Bajjurah.

APPENDIX D

THE QUARRIED VINEYARD – AN UNKNOWN AGRICULTURAL PRACTICE FROM THE ROMAN-BYZANTINE PERIOD¹

Adam Zertal

INTRODUCTION

Beginning in the 80s of the 20th Century and ever since, concentrations of deep rectangular shafts, quarried in the rock surfaces, have been found scattered over vast areas of the survey. Roughly, the shaft fields or groups are spread over the area between the Sanur – er-Rammeh line (south), Nahal 'Iron (north), the Sharon plain (west), and the fringes of the Jezreel Valley (east).

Several explanations for these have been put forward: one was that they were cemeteries of the Intermediate Bronze period (Mazar 1990: 159-162); however, their numbers and distribution invalidated this and other suggestions.

After a process of elimination, it now seems that these shafts represent a previously unknown method of growing vines on rock surfaces, by using large uncultivated areas of bare rocks. We call this method the 'quarried vineyard'. It adds new information to the ever-growing body of knowledge of Roman-Byzantine agriculture (Felix 1963; Dalman 1964: 294-413; Dar 1982; Frankel 1984).

THE REGION'S FORMATION; GEOLOGY, CLIMATIC AND SOIL DATA

The northern hill-country of Canaan/Israel (Samaria region) is built of three main geological formations: the Shechem syncline, the Far'ah anticline, and the Umm el-Faḥm anticline. All three exhibit large surfaces of hard and soft chalky rock, parts of which have been fully exposed as their soil cover has been washed away by rain and erosion.

Considerable sections of the hilly area have been under cultivation since the Bronze Age. The main technique for producing arable soil here is the terrace, which support parts of the slope and collects soil. However,

¹ This article is a short version of Zertal 1999. The late Nivi Mirkam and David Eitam participated in crystallizing the ideas expressed here. Thanks are due to M. Brody from Alon Moreh for bringing the quotations from the Mishnah to my attention.

in large sections of the hills the steep slopes and the summits remained as rocky exposures with no cultivation.

The limestone and chalk layers are full of fissures and cavities which split the rocks in all directions and create pockets in them. The 'quarried vineyard' builders took advantage of these pockets, filled with washed-in or imported soil, and they became the basis for the vines. The square shafts were intended to connect the outer surfaces with these pockets.

The main soil here is *terra rossa*, a reddish, relatively fertile soil which is the product of the erosion of limestone. With a proper supply of water, this soil well suits the Mediterranean horticulture of olives, vines, etc. A further advantage of the method discussed here is the solution of the water prob-

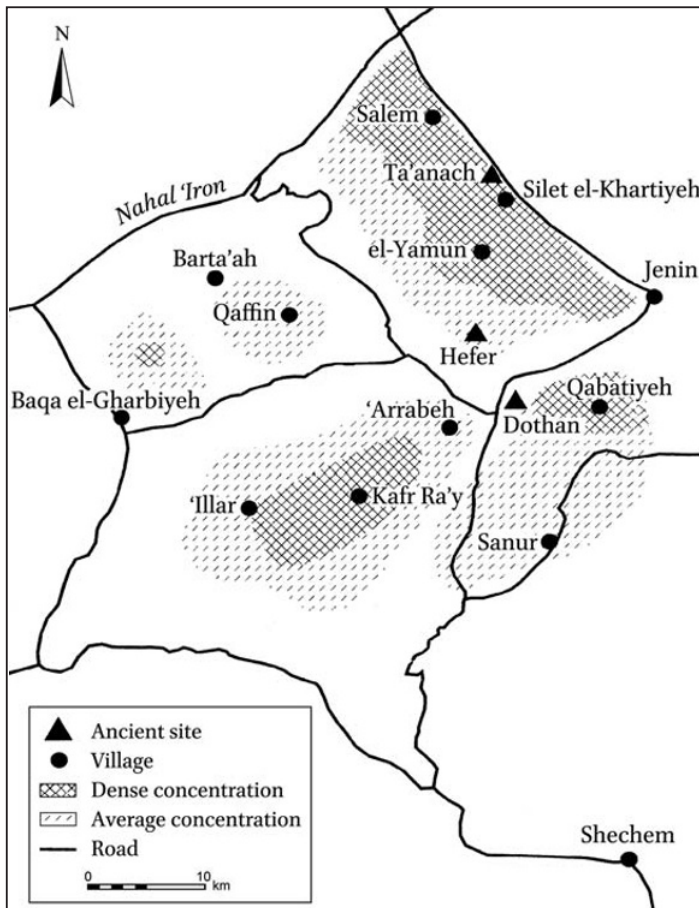


Fig. D1: Northern Samaria and the main locations of quarried vineyards.

lem by concentrating the rainwater and directing it to the plants.

The Mediterranean climate, with long dry summers and short winters, and with 500-600 mm annual rainfall, also suits the quarried vineyard, by conserving the rainwater in the soil pockets during the summer, where it is available to vines or trees.

THE SYSTEM (FIGS. D2-3)

The vineyard consists of four elements:

- Square shafts;
- Cupmarks;
- Water channels, and
- Winepresses with their equipment.

The **square shafts** are each 50×50 cm on the average, and 1 m or more in depth. Their distribution on the surface is irregular, mostly located in rock depressions for easy water supply. The distance between the shafts averages 2–3 m. Today they are full of washed-in soil, in many cases still containing a fig or olive tree. The groups of shafts are concentrated on steep slopes or summits, and they seem to represent private or group ownerships, like regular vineyards. In or near some of the groups are quarried winepresses.

The **cupmarks** are small depressions carved in the rock, 10 cm in diameter and 5 cm deep on the average. They are scattered around each shaft, 4 to 7 in number. They were most probably used as bases of wooden trellises, used to hold the vines above the hot rocks.

Channels: In many cases, but not always, shallow channels have been cut in the rock to direct the rainwater into the shafts.

Winepresses for wine production are an organic part of the project, having been found near many of the square shaft groups.

These elements together comprised a vineyard unit. The square shafts are grouped in concentrations, which probably represent their owners.

Why a quarried vineyard? The clear advantages of the system are:

- Large areas of uncultivated sterile rocks became fertile fields;
- The plant enjoyed better conditions than usual. Above all, the quantity of water collected was much greater than that of normal vines;
- There was no need to clear weeds, little water was lost by evaporation, and all food and water is directed to the plant.

However, much work was needed to build the system. A team of stone-

workers from a village near Jenin has estimated that one shaft would take 3 to 6 hours of quarrying, depending on its depth. This means that an average hill (100 shafts) required 300 to 600 working hours, a figure to which the cutting of the cupmarks and the other elements should be added.

Nonetheless, the advantages here outweighed the disadvantages, and the return for a relatively small investment seems to be a good one.

THE QUARRIED VINEYARD IN THE TALMUDIC SOURCES

Following the discovery, it was found that the Talmudic literature from the 2nd Temple period and later was aware of the method. Thus, in the Mishnah (Kila'im 5, 4) is written: "A vine planted in a winepress or a shaft, is to be treated regularly and cultivate the rest; Rabbi Yossi says: If there is not 4 cubits there it is not to be cultivated" and in Kila'im 5, 3: "The winepress in the vineyard is 10 cubits deep and 4 wide. Rabbi Eliezer says: you should cultivate inside".

'Winepress' here means any cutting or quarrying in the rock, and also

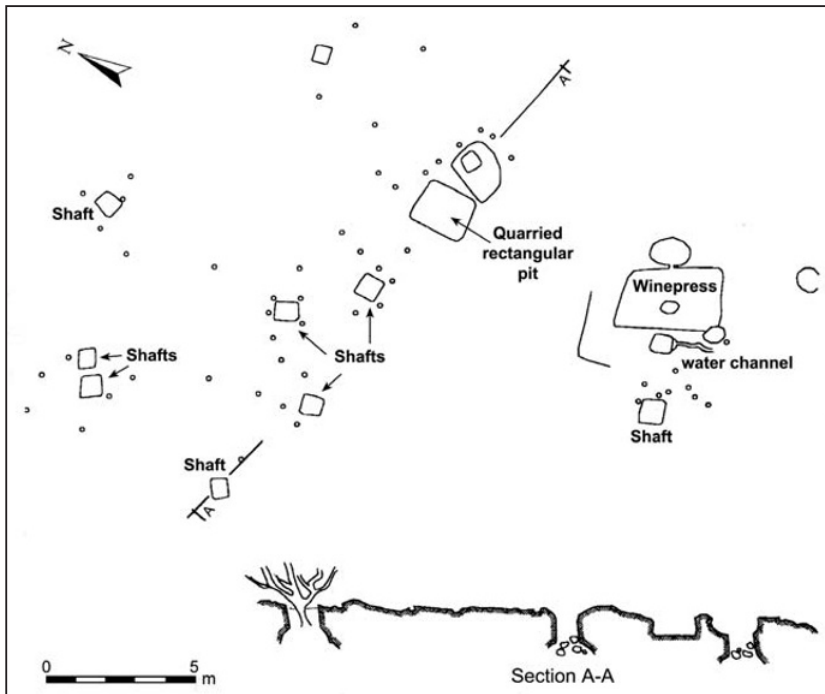


Fig. D2: Plan and section of a quarried vineyard east of the village of Salem.

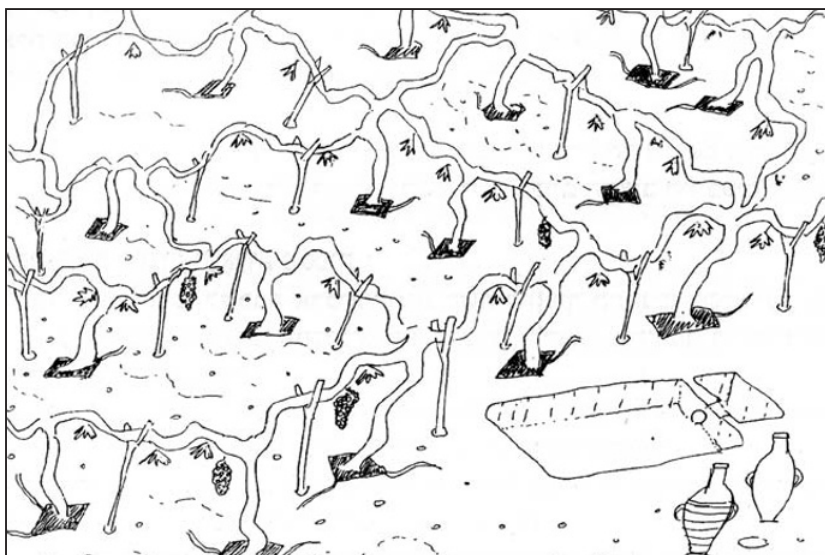


Fig. D3: Reconstruction of a quarried vineyard.

includes the square shafts, called '*neqa*' in Hebrew – a term which should be applied to the square shafts. The Even-Shoshan definition (1966, IV, 1722) is "Crack, hole in the ground" and quotes the phrase from Bava Batra 7, 1: "Neqai'm which are 10 tefahim deep", which is 2 m deep and suits our shafts.

As for raising the vines upon trellises, the Mishnah again (Kil'aim 6, 3) says: "The one who raise the vines upon 'apifiarot' will not cultivate under the permitted. If he did – it is not holy; and if the new one had gone – it is unpermitted".

This difficult 'halacha' speaks of a quarried vineyard, using the rare term 'apifiarot' in Hebrew, which derives presumably from 'apirion' – a light bed carried by slaves, used to move kings or nobles from place to place. 'Api-fiarot' could be the standing wooden trellises (Even-Shoshan 1966, I, 137).

DATING AND OTHER PROBLEMS

Dating of agricultural installations is difficult, due to the lack of stratigraphy and the conservatism typical of agricultural methods. Nevertheless, we would date the quarried vineyard to the Roman-Byzantine period, based on the following considerations:

- This period was the peak of agricultural expansion in Palestine, due to

- the ever-growing population, peace, and security;
- The Roman-Byzantine settlement was by far the largest in the area;
- The regular sherds around the shafts are from these periods (Roman and Byzantine);
- All the winepresses related are of Roman-Byzantine types (Dar 1982: 230-268; Frankel 1984: 204-211).

Vine-growing has been well treated by Dar, Frankel (1984), and others. Roman literature indicates one winepress to 80 dunams (8 ha) of vineyard, while Dar (1982: 230-268) suggests 45-60 dunams (4.5-6 ha). Ahlström (1978: 45-46) reports 117 winepresses in the Ta'anach region, all dated to the Roman-Byzantine period, which may represent about 8000 dunams (80 ha) of vineyards.

The new discovery, however, raises problems. Firstly, why is the method not found elsewhere in the hill country of Israel/Palestine? Secondly, what is the reason for it not being known throughout the Roman Empire? The 'quarried vineyard' seems to be so effective that one should assume its being a basic practice across the ancient world, which seems not to be the case.

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APPENDIX E

A BRONZE FIGURINE FROM KHIRBET YANNUN (SITE 42)

Adam Zertal

During the survey at Khirbet Yannun (grid 1727/2121) near Tel Ta'anach a bronze figurine was found (Fig. E1), an addition to the ever-growing collection of these small finds from the 2nd millennium BCE.

The figurine, 116 mm tall, was cast by the 'lost wax' technique. It represents a sitting male figure with forearms extended forward from the elbows and fists clenched in a grasping position. The head details are well-worked: the nose is big and convex; the mouth wide and the ears protruding. The deep eye sockets were probably originally inlaid. The face has a long and styled beard, and the figure wears a tall cylindrical tiara on the head. The tiara is adorned with three pairs of horns, a divine symbol, with a fan-like crown on the top. Contrary to the head, the body is quite rough and lacks details.

The figure is clothed in a long gown with a broad hem, and the soles of the feet are well worked with sandal straps. The broad bronze wedge protruding downwards from the feet was used to fix the figurine to a wooden pedestal.

According to the classification of Negbi (1976), the figurine belongs to the 'Enthroned Deities and Worshipers' group, part of the 'Reigning God' general type. In most cases, and unlike ours, the arms of the reigning gods are extended or raised in a blessing gesture. Sometimes the deity is holding an object in its hand. Generally, the Reigning God type is rarer than the second group – the Warring Gods.

Geographically, our figurine belongs to the second Syrian-Anatolian group, common in north Syria, Anatolia, Cyprus, and Canaan during the 2nd millennium BCE. Greatly influenced by Mesopotamian and Anatolian motives, the long beard and radiating tiara of the figurine are deity symbols in reliefs and cylinder seals in Mesopotamia from the 3rd millennium BCE onwards: on the other hand, the long broadly hemmed gown is a Syrian-Anatolian tradition. The body as a flat bronze plate is distinctive to the Syrian figurines originating in Anatolia, and it seems likely that 'the Enthroned Reigning God' originated in the Mesopotamian 'Presentation Scene', where the god accepts his subjects' gifts.

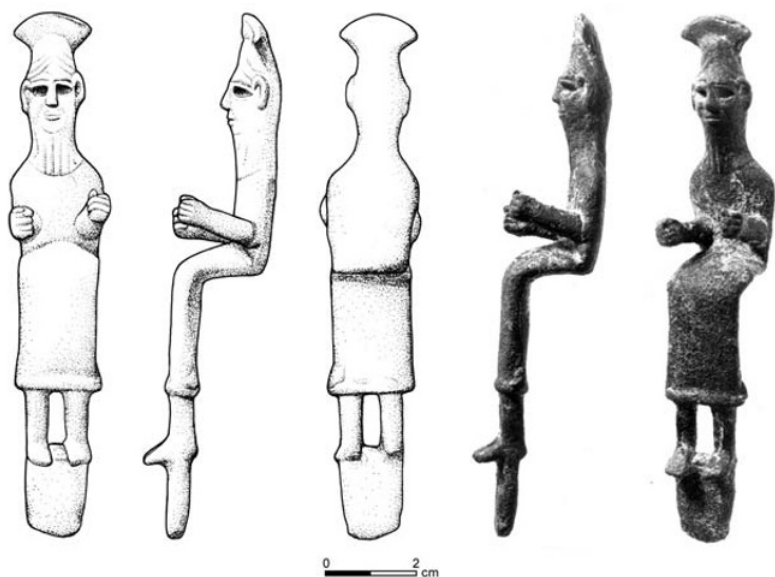


Fig. E1: The Bronze Figurine from Khirbet Yannun.

The uniqueness of this figurine is the mixture of styles. The crown adorning the high tiara should be the plumed crown which appears as a special headdress in a group of 'Warrior' figurines originating in northern Syria.

Porada, in an important article on the subject (1942), showed that warriors with plumed crowns, named by her 'Syrian-Cappadocian', appear on cylinder seals from Cappadocia from the First Dynasty of Babylon (20th–19th centuries BCE) onwards. These unique crowns also appear on figurines from Ugarit, dated by Schaeffer to the 19th–17th centuries BCE, or the Middle Bronze Age IIa. A third group of Warrior figurines with plumed crowns comes from Tell Simiryan south of Ugarit and other tells in the Orontes Valley in Syria. Negbi (1961; 1970) also relates this group to the Middle Bronze Age IIa. In spite of the fact that two other Warrior God figurines may possibly be related to the Late Bronze Age, the majority of the plume-crowned warrior figurines are to be dated to the first half of the 2nd millennium BCE.

Our figurine, therefore, represents a combination of artistic elements from various periods: the plumed crown is typical of Warrior images of the Middle Bronze Age, but in this figurine it adorns the head of a 'Reigning God'. The 'Enthroned Reigning God' is very characteristic of the Late Bronze Age, and such items have been found both in Syria – like the Kistna

figurine – and in Israel. Hansen (1957), who investigated the subject, has followed the evolution of the 'Flat Idol', from the warrior with the plumed crown to the 'Enthroned God'. According to him, there is still a missing link between the Middle Bronze Age figurines and those of the Late Bronze Age.

It can be suggested that our figurine represents this missing link. It seems that the plumed crown is an addition to define a Reigning God as a warrior. No parallels have so far been found, but it can be assumed that such group did exist. If the assumption is correct, then our knowledge regarding the religion and idols of the inhabitants of Canaan in those ages has widened a little.

The archaeological context of the figurine could help. Khirbet Yannun is located at the fringes of the Jezreel Valley, about 2 km south of Tell Ta'anach. Tombstones that were shifted from their position during agricultural cultivation and fragments of clay burial coffins, which were discovered during the survey, point to a large cemetery. The sherds included both Middle and Late Bronze Age examples. Vessels of the Bichrome type (including a shard with an image of a bird) and 'Chocolate on White' ware may be dated to the transition period of the MBIIc-LB I (16th century BCE).

It is difficult to know if the figurine was made by a local artisan, or was imported from northern Syria. The latter possibility is more reasonable; and if so, it might point to the diverse connections between Canaan and Syria at the time.

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APPENDIX F

PRELIMINARY REPORT ON THE 2010-2012 SEASONS OF THE EXCAVATIONS AT TEL ESUR (ASSAWIR [SITE 30])

Shay Bar

INTRODUCTION

The site of Tel Esur (Site 30; in Arabic: Assawir) is located in the north-eastern Sharon Plain, standing strategically near the western entrance to Nahal 'Iron (Wadi 'Ara), 18 km south-west of Tel Megiddo (Fig. F1).

Two springs flow from several outlets in the fields surrounding the tell: the eastern springs ('Ein Arubboth or 'Ein Assawir) 100 m to the east, and the western springs 50 m to the south-west. Nahal 'Iron flows 800 m south-east of the site.

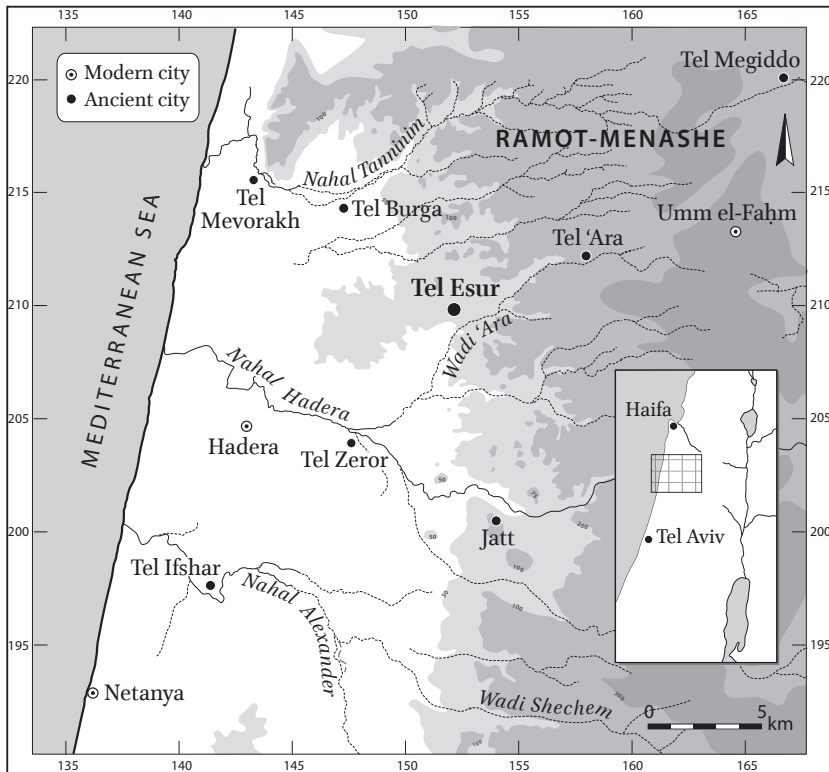


Fig. F1: Tel Esur - general location map.

The upper area of the tell is 2.5 ha, with two additional sites in its vicinity: a smaller tell, about 300 sq. m in area, 100 m east of the tell (area D in Fig. F2); and a very large protohistoric site about 7 ha in area, lower and to the south of the tells. Together they constitute one large site (Yannai 2006) (Fig. F2).

Tel Esur was first surveyed by the Palestine Exploration Fund, and from then on has been intensively surveyed by various teams (for a detailed description of surveys see Site 30 in this volume, and Yannai 2006: 5-6; for the different geographical-historical identifications see this Volume,

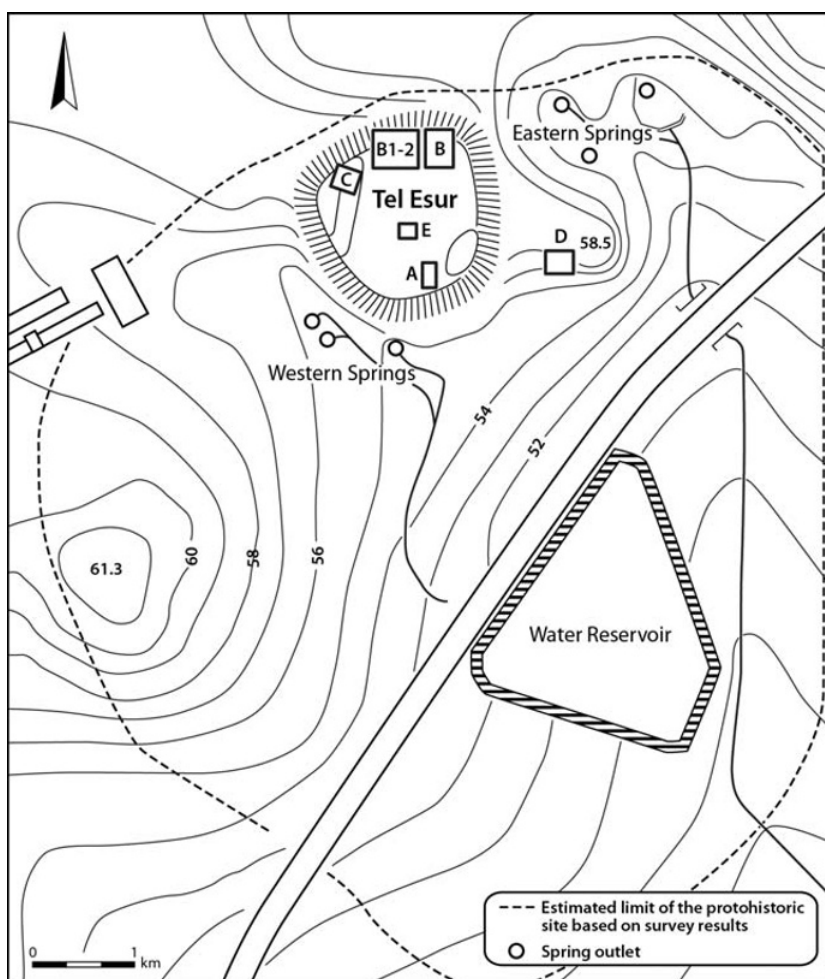


Fig. F2: The Tel Esur area (after Yannai 2006: fig. 1.2).

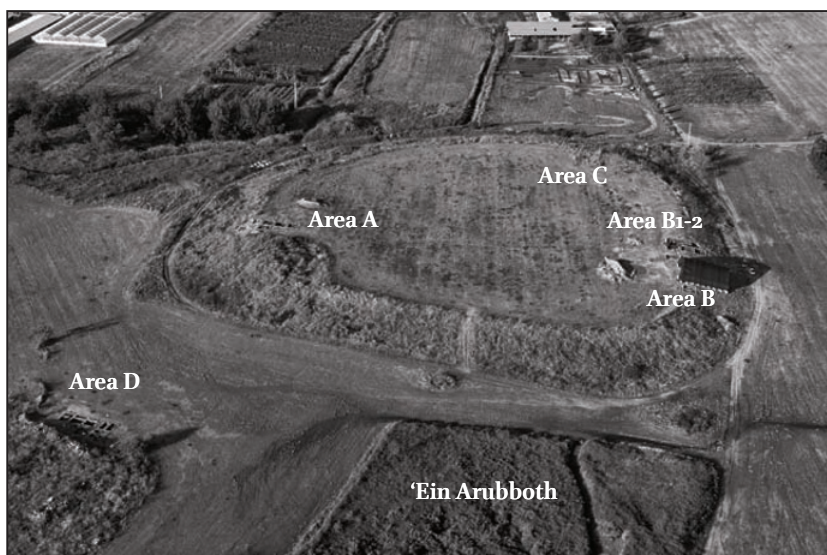


Fig. F3: Aerial view of Tel Esur and the main excavation areas, looking west.

Chapter 2). All the investigators reached the same conclusions: the site was first settled in the Middle Bronze Age II, continued to exist in the Late Bronze Age, and from that time on there was sporadic settlement in the Iron, Persian, and Roman periods.

The place was probed by Zertal (2003) between 2001 and 2003, and the major discovery was the Middle Bronze Age fortification system in Area B, the northern part of the tell. A section in the north-eastern part of the tell (Area C) exposed a Late Persian/Early Hellenistic stratum above a fill dated to the Late Bronze Age with no architectural remains, which was in turn on top of Middle Bronze Age domestic walls.

In the renewed excavations undertaken by the Tel Esur Community Project, inaugurated in 2010, the excavation was much enlarged. Four main areas were opened in different parts of the tell and its immediate surroundings.

AREA A

Area A is situated in the slightly higher southern part of the site. The place was heavily damaged in modern times, preventing a clear stratigraphic sequencing. In the 150 sq. m excavated the remains of eight phases were exposed (from top to bottom):

Phases A₁ and A₂

Remains of modern trenches, pits and ancient walls, heavily damaged by military trenches which cannot be dated. Since the latest pottery recovered dates to the Roman period, it may indicate the last of these phases.

Phase A₃

The remains of several walls (W₄₁₀₀₉, W₄₁₀₃₆, W₄₁₀₂₆, W₄₁₀₅₇, W₄₁₀₄₆ and W₄₁₀₂₃; Fig. F4) and three pavements (L₄₁₀₂₇, L₄₁₀₃₈₋₃₉ and L₄₁₀₅₈) were found. These have also been damaged by modern activity, but some restorable pottery (a complete cooking pot, two jars, a Herodian lamp, and a decanter), found on floors abutting the walls, have dated this Phase to the Early Roman period.

Phase A₄

Damaged remains of two walls (W₄₁₀₅₅ and W₅₁₀₃₀) and a surface with a tabun (L₄₁₀₄₁), dating to the 1st century CE were found (Fig. F4).

Phase A₅

A pit dating to the Iron Age IIb was found in Square V10. This is the only evidence of Iron Age II activity found in the renewed excavations of the tell.

Phases A₆ to A₈

Three phases were exposed in Squares V9-10 and U11 (Fig. F5): Phase A₆ includes partial stone surfaces (L₅₁₀₅₅₋₅₆, L₆₁₀₂₉, and L₆₁₀₂₈ in Fig. F5, and not shown in the figure: L₅₁₀₃₅₋₃₆ in Square U/11 above W₆₁₀₂₆ of Phase F8), and a wall (W₅₁₀₆₀); Phase A₇ includes a stone surface (L₆₁₀₃₀) and wall (W₆₁₀₂₇); Phase A₈ includes walls discovered in Square U/11. All these are dated to the Middle Bronze Age IIb-c.

The excavation in Area A, in spite of the modern destruction, has determined some basic factors regarding Tel Esur:

1. This is the only area where evidence of Early Roman occupation has been exposed. From the limited exposure and fragmentary remains it can be deduced that a few buildings, perhaps a farmstead, stood in the southern part of the tell in this period.
2. The pit in Square V10, with Iron Age II activity, can possibly be related to the large Iron Age building excavated in the lower tell (Area D – see below).

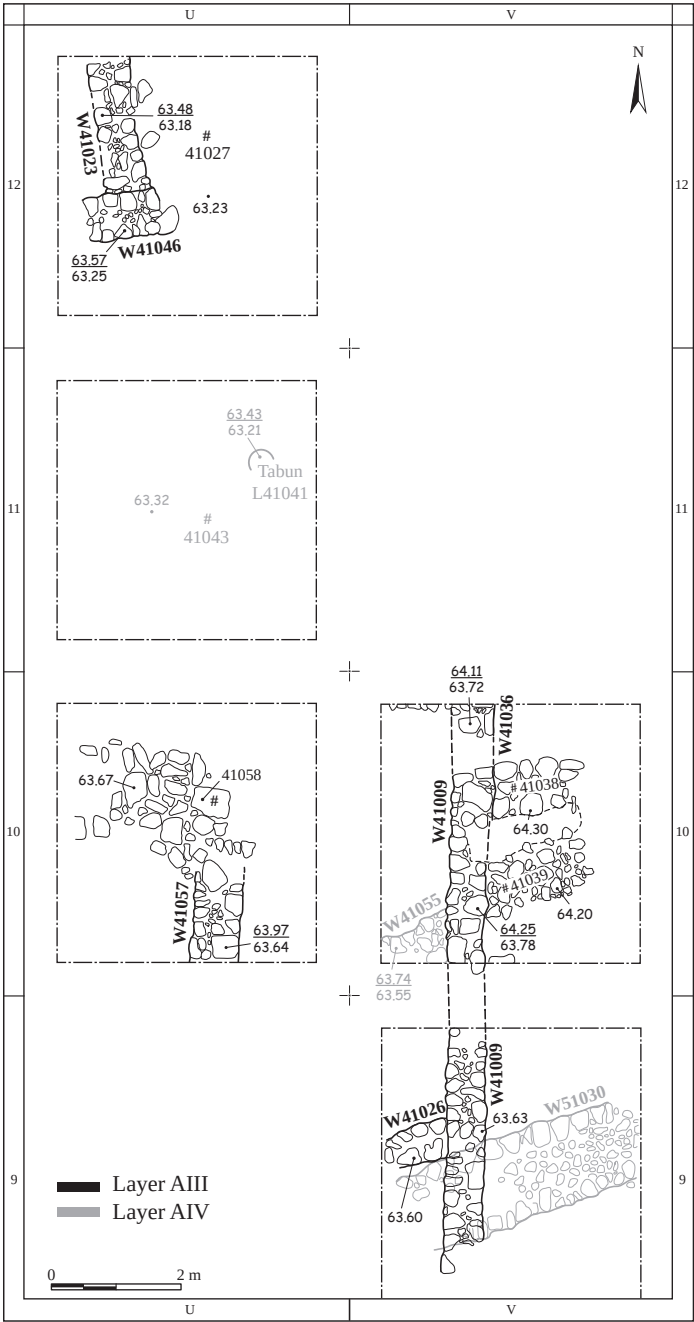


Fig. F4: Plan of Area A, Phases A3 and A4.

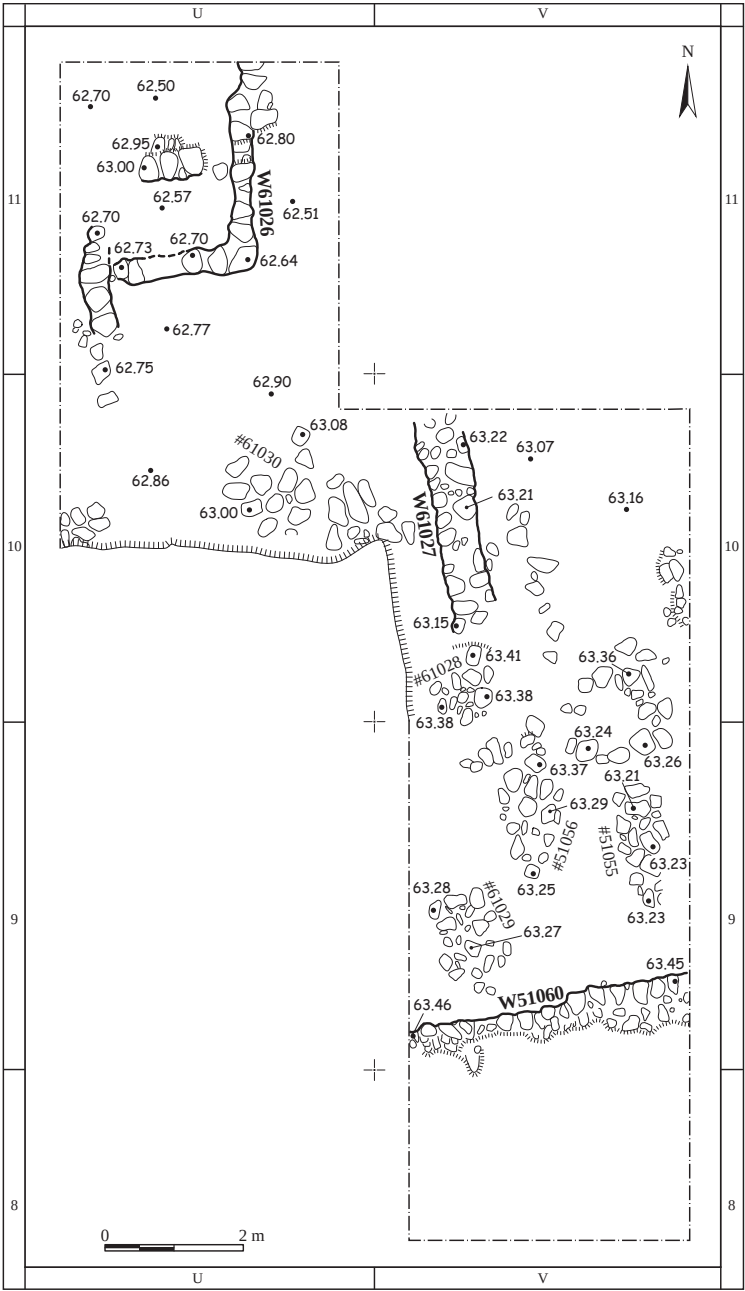


Fig. F5: Plan of Area A, Phases A6 (pavements in Squares U/10 and V/9-10), A7 (Wall 61027 and surface 61030), and Phase A8 (walls in Square U/11).

In the renewed excavation the area was expanded to search for further elements. The 'glacis' was found to be a unique pyramidal-shaped tower, leaning against the city wall and abutted by a ramp leading into the city (Fig. F6). This type of construction is unparalleled so far.

Three strata have been discovered in the 250 sq. m excavated. These are, from top to bottom:

Stratum B1

The scanty remains of walls on the surface and slope of the tell belonging to more than a single cultural horizon. These include the following:

- The 2001 'Roman terrace' on the slope of the tell. This 30-m (minimum length) wall was roughly built of medium and large stones, and encir-



Fig. F7: Stratum B2 building (shaded) in Area B.

cles the lower slope of the tell in Areas B and B2. The Late Roman body sherds collected beside it suggest an agricultural element, preventing the soil from washing off the tell;

- A 1 m-thick Late Bronze Age wall in Square Y/42. The remains of a beaten earth floor abutting a well-built wall with fine masonry were discovered close to the surface level. The pottery assemblage is Late Bronze Age, and the structure that stood here may be connected to the Late Bronze Age strata discovered in the adjacent Area B1;
- Remains of two adjacent walls in Square X/42. These fragmentary walls, discovered close to the surface, are to be dated to a post-Middle Bronze Age phase in this area, but their exact dating remains obscure.

Stratum B2

A well preserved Middle Bronze Age IIb building was excavated in Squares V and W/42 (Figs. F7, F8). Its known dimensions are 6 x 10.5 m (enclosed by walls W12008, W22030 and W62035), with the western closing wall missing. The walls still stand to more than 1 m high. A thinner northern wall with different masonry (W62035), was built on top of the Middle Bronze Age IIA city wall W12009.



Fig. F8: Stratum B2 walls superimposed over Stratum B3 walls and partly visible remains of the city wall, in Area B, Sq. W/42, looking north.

The building was divided on the east by the upper part of wall W22008 (Fig. F8). The remains of large pithoi body sherds were found *in situ* on the building's crushed lime pavement.

Stratum B₃

The Middle Bronze Age fortification system was revealed here. A unique pyramidal tower with three oblique walls (W12017 and W42012) was excavated in Area B, and W42620 in Area B2 (Fig. F9). It was built against a thick stone wall (W12009 – Fig. F10), its base measuring approximately 6×11 m. The tower had a homogeneous fill, on a stone floor (F42026) that probably served solely as an architectural foundation. The upper floor of the tower, the observation level, did not survive, and should have been higher than the surface of the tell.

The sealed loci below the stone floor (L42027) have revealed additional courses of the city wall: the foundation has not yet been reached. No parallels have so far been found in Canaan or Syria to this impressive tower, and both the pottery from the fill of the tower, and the collected pottery from sealed L42027, were dated to the Chalcolithic, Early Bronze Age I, and Middle Bronze Age IIa, thus giving a *terminus post quem* for the tower in the Middle Bronze Age IIa.

Another feature of the fortifications is the earthen ramp built against the city wall at a very obtuse angle. This comprises loose brown soil interspersed with patches of crushed limestone and field stones (Fig. F11), incorporated into the fill to prevent erosion. A stone pavement (L22023 and L52033) abutting the city wall and the tower base was revealed at the bottom of the earthen ramp. It served as rubble bedding for a thick mud-brick wall built against the stone city wall, and in turn was abutted by the ramp fill and its limestone patches. The mud-brick wall was probably constructed to reinforce the wall in this location so that it could support the ramp.

The fill and limestone patches stand against the eastern wall of the tower (W12017), not appearing on the western side (W42620 in Area B2), while the city wall did extend beyond it. A possible explanation for this, supported by the moderate ramp, is that east of the tower was part of a gateway complex, or that the ramp was built to support the city wall only in this location for constructional reasons.

On the lower courses of the inner face of the city wall we unearthed two thin well-preserved parallel walls (W32014 and the lower part of W22008),



Fig. F9: Stratum B3 tower in Area B, Sq. W/44, looking west.

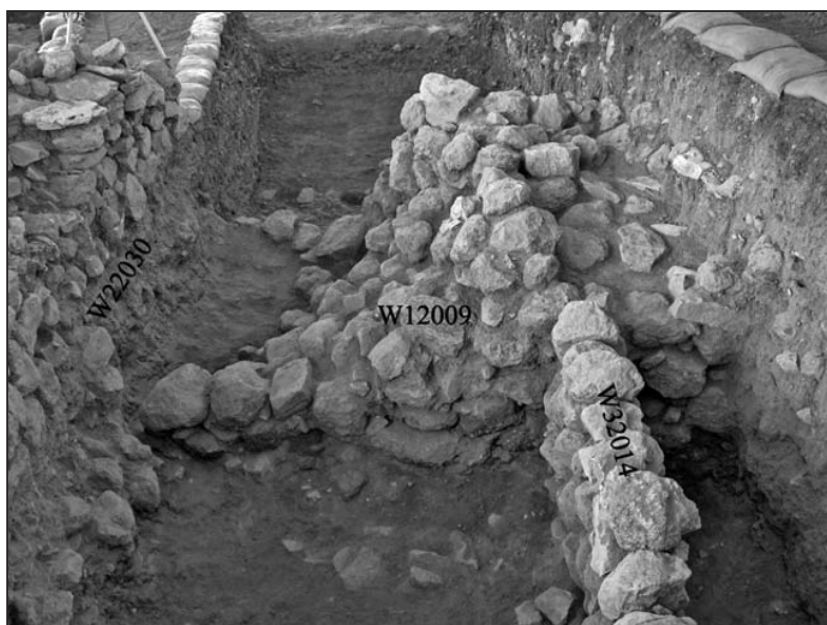


Fig. F10: Stratum B3 robbed city wall at Area B, Sq. X/42. Note the thin wall abutting the city wall from the inside, looking north-west.

but have not yet reached their bases.

Surprisingly, no chronological difference was discerned between the pottery assemblages of Strata B2 and B3 (Fig. F12). Many of the types here debut in early Middle Bronze Age IIA assemblages in other Sharon sites, such as the 'hammer-head' jug rim (Fig. F12: 10, 13) and the S-shaped bowl rim (Fig. F12: 1) known in Aphek Phase 2 ['pre-Palace']. For jug rims see, e.g., Beck 2000: fig. 8. 12: 16 [Str. B Vc]; for S-shaped bowl rims see, e.g., figs. 8.10: 5 [Str. B Vd]; 8.12: 6–7 [Str. B Vc]. However, there are several types that strongly point to transitional Middle Bronze Age IIA-b and early Middle Bronze Age IIB dates. The shallow open bowl with an everted rim (Fig. F12: 4) is a dominant type in Aphek in the Middle Bronze Age IIB (e.g., Yadin 2009: fig. 7.14: 9, 11–19), also considered a Middle Bronze Age IIC type at Shechem (Cole 1984). The types appearing at Aphek are usually not slipped, and occasionally have a slipped rim, whereas our example is slipped on the rim and inside.

Based on the preliminary data, we suggest that the earliest excavated Middle Bronze Age levels at Tel Esur are from the Middle Bronze Age IIA. Unlike other fortified Sharon Plain sites, mostly abandoned in the second stage of the Middle Bronze Age, Esur flourished in the Middle Bronze Age IIB. Although its defence system may have gone out of use during this



Fig. F11: Stratum B3 city wall and external features in Area B, Sq. X/42-43, looking south-east. Note the moderate ramp slope in the section.

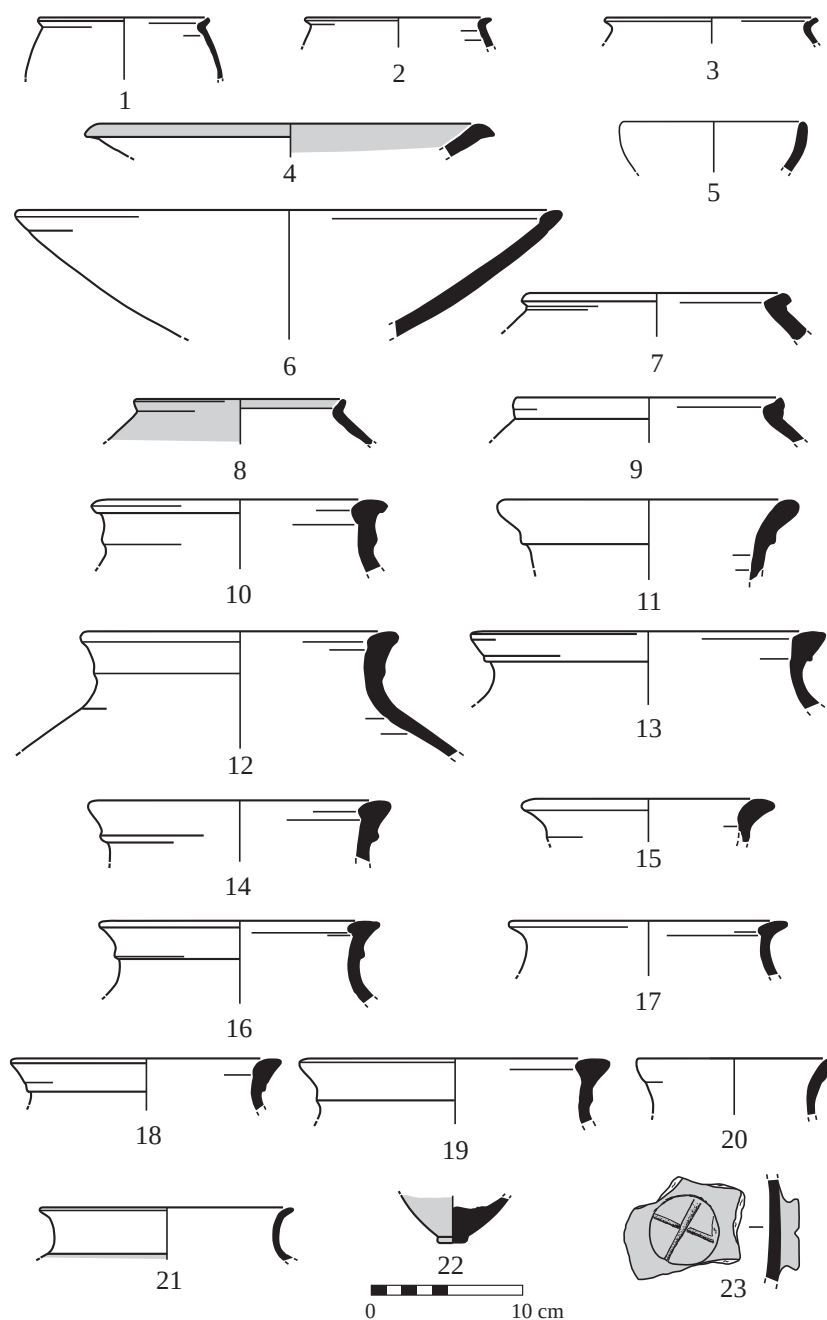


Fig. F12: Selected examples of Area B Middle Bronze Age II pottery.

period, large-scale buildings were still being erected, and pottery production continued. It raises the question of the difference between Esur and fortified tells like Poleg, Zeror and Burga. The answer probably lies in Esur's affinity with the powerful site of Megiddo and the 'Arunah pass.

AREA B1

Area B1 is situated on the northern slope, adjacent to Area B2 and close to Area B. The major feature here is a Late Bronze Age IIA structure (Figs. F13 and F14), located in Sqs. R-T/39-42. The structure consists of a presumed main courtyard (L52203), with rooms abutting it, some with restorable pottery. The preservation of the structure is fairly good, apart from the upper 10–15 cm, which were ploughed in modern times. The limits of the building are not fully clear, since some of the walls were cut by post-depositional activities (W62280 to the south; W42214 and W52220 to the north; and W62250 and W52218 to the east). Only the foundations were

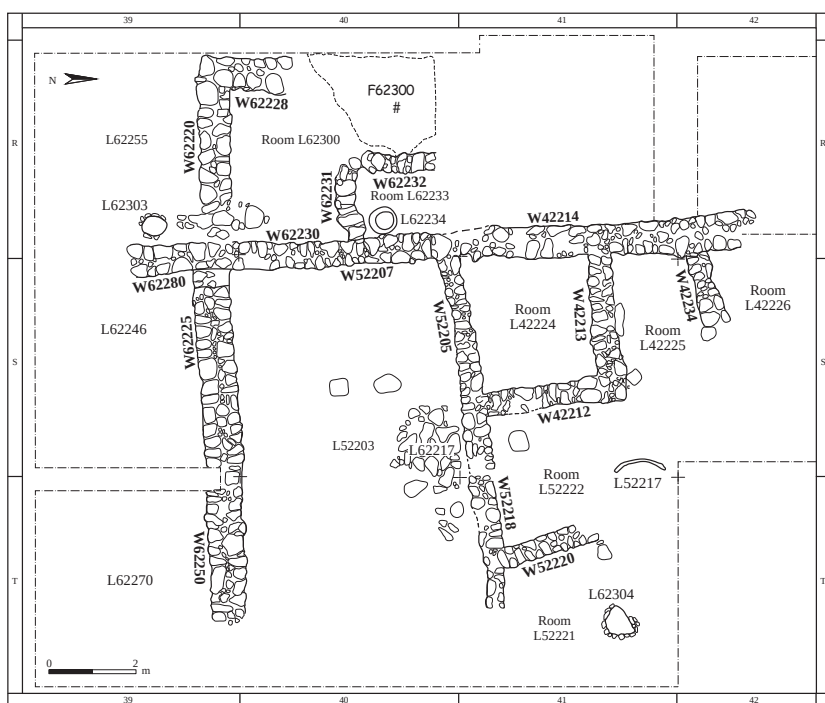


Fig. F13: The main architectural features of the Late Bronze Age structure in Area B1.



Fig. F14: Aerial photo of Areas B1 and B2 at the end of season 2012.

preserved, and the presence of mud-brick material and collapsed mud-bricks suggests a mud-brick superstructure.

The main restorable pottery assemblages originated from the following deposits:

- Room L42224 located in Square S/41 had at least six restorable vessels, including four large pithoi (e.g. Fig. F15). This room was used for large storage vessels, and a royal scarab of Amenhotep III was found in the surface of the square (Fig. F16).
- Room L42225 in Square S/41 included at least seven restorable jars (Figs. F17 and F18). This room, adjacent to room L42224, was also used for storage, with slightly smaller containers.
- Room L62233 in Square R/40 contained a *tabun* (L62234, Fig. F19), two restorable vessels, and a female plaque figurine. This small room was used as a cooking area or kitchen.
- L62270 in Square T/39 included seven restorable vessels. Since most of L62270 is located below the floating level of W62250, it is not clear whether the pottery belongs to the same phase as the large building. This Late Bronze Age assemblage may be earlier than the architecture and assemblages collected in the other rooms of the building, so it may



Fig. F15: Restored pithos from room L42224 in Area B1.



Fig. F16: Royal scarab of Amenhotep III from Area B1.



Fig. F17: Room L42225 during excavation, looking south.



Fig. F18: Restored Late Bronze Age jars from room L42225.

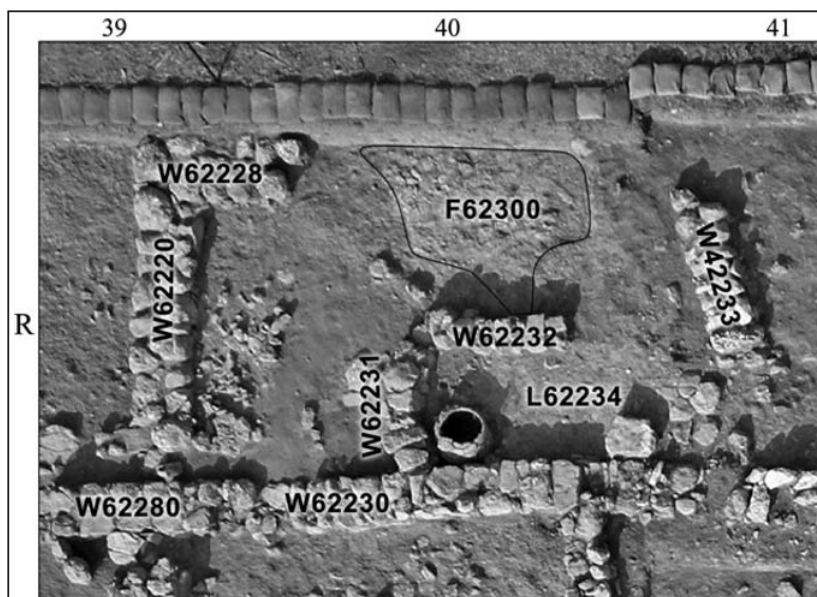


Fig. F19: The *tabun* in Sq. R/40, looking west.

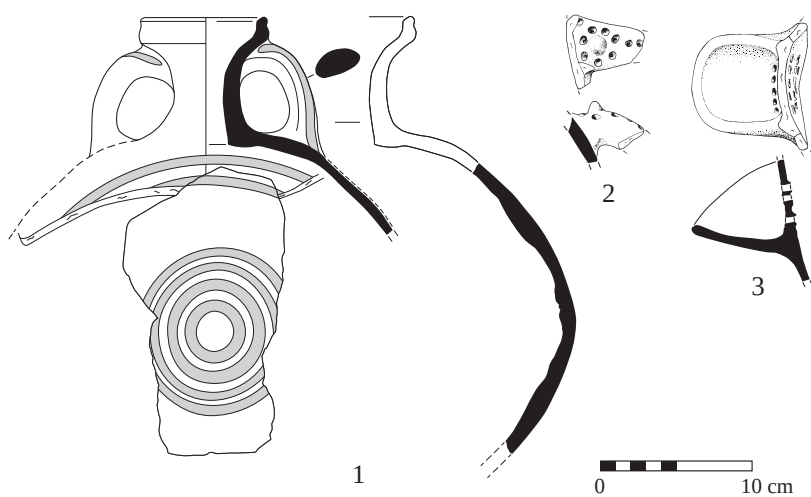


Fig. F20: Pottery of the Iron Age I pit in Area B1.

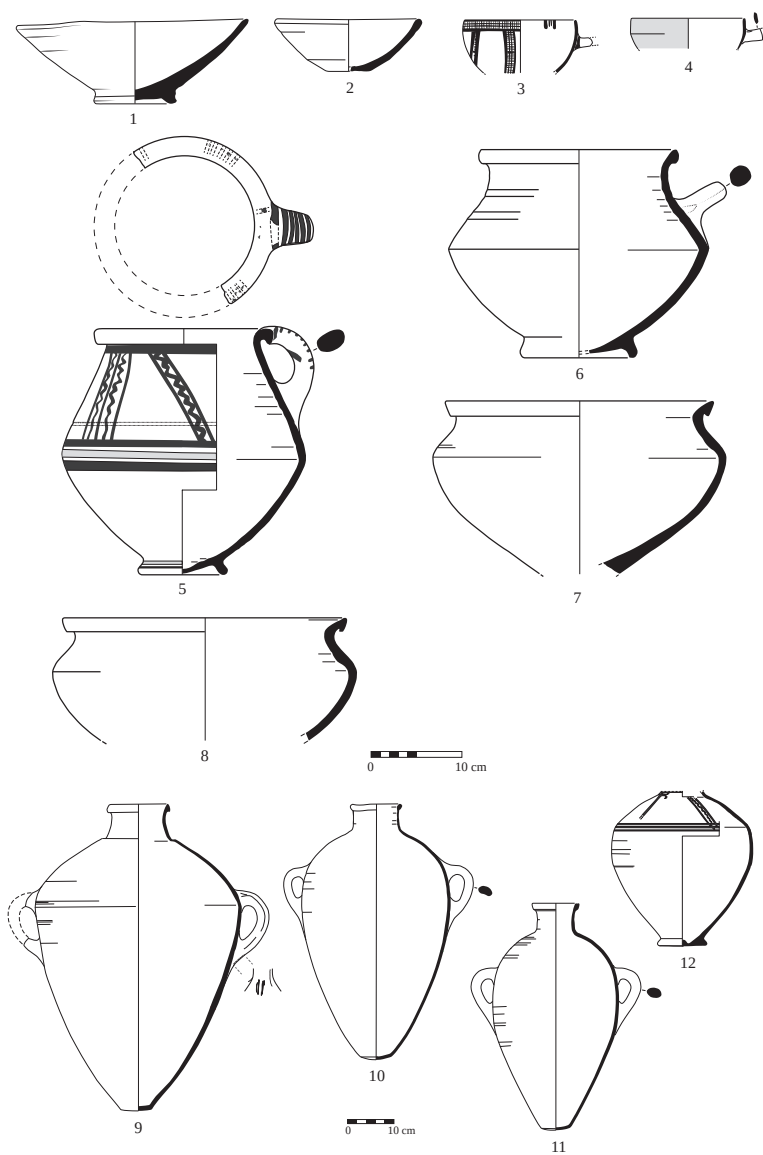


Fig. F21: Selected examples of Area B1 pottery.